

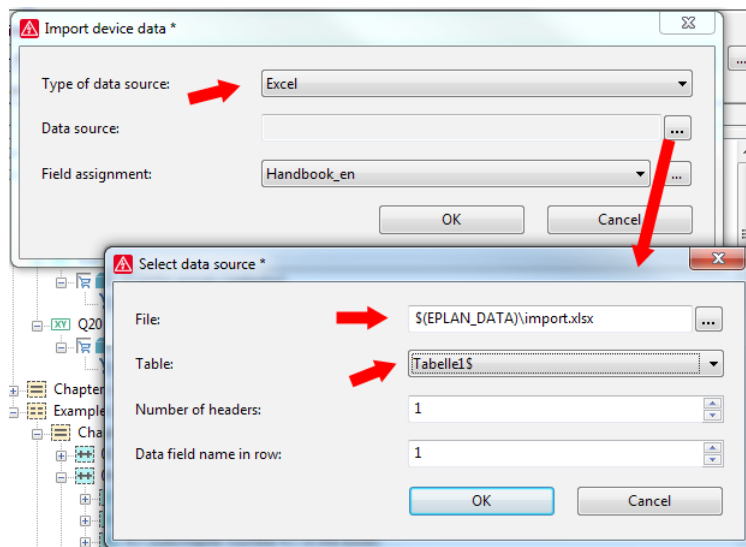


Fig. 5.47 Configuring import data

Once done, the dialog is confirmed with OK. EPLAN imports and opens the **SYNCHRONIZE DEVICES** dialog. Changes can be made in this dialog.

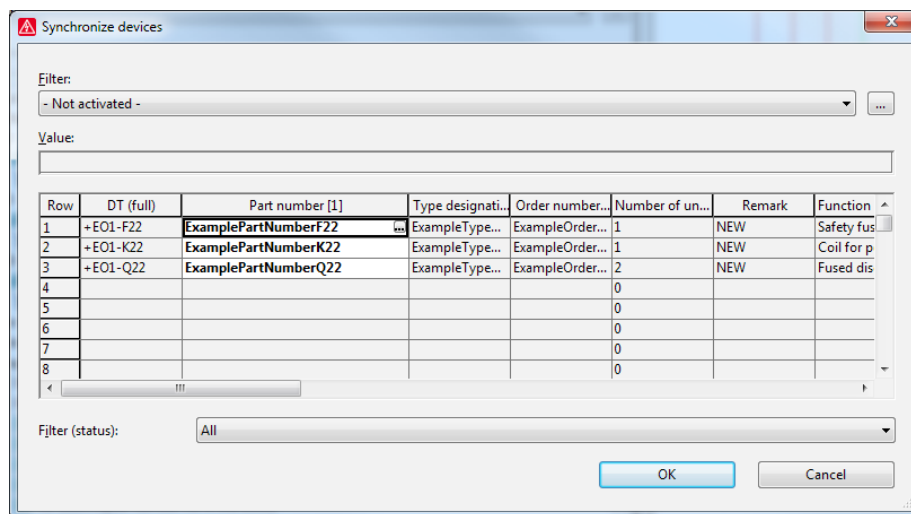


Fig. 5.48 Synchronize devices dialog

After clicking OK, EPLAN applies any changes made, imports the data, and displays a message about the successful import.

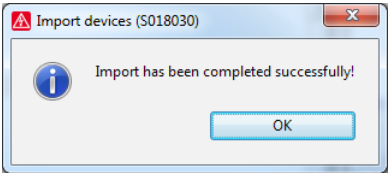


Fig. 5.49
Message

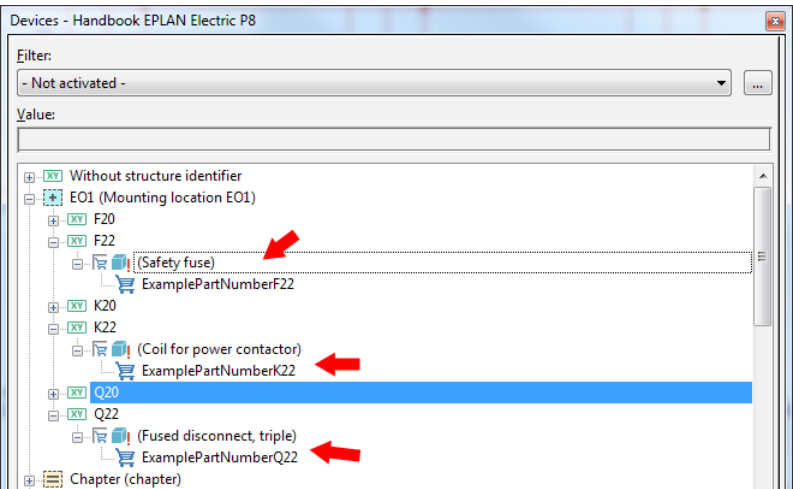


Fig. 5.50 Imported device data

5.2.7 New

The NEW... function inserts a new function definition, for example an additional auxiliary NC contact for a motor overload switch.

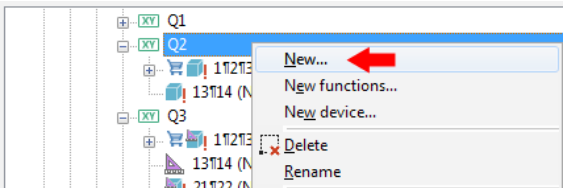


Fig. 5.51
New... menu call



First you select the motor overload switch in the device navigator. Then you open the popup menu in the device navigator and select the NEW... function. EPLAN opens the FUNCTION DEFINITIONS dialog.

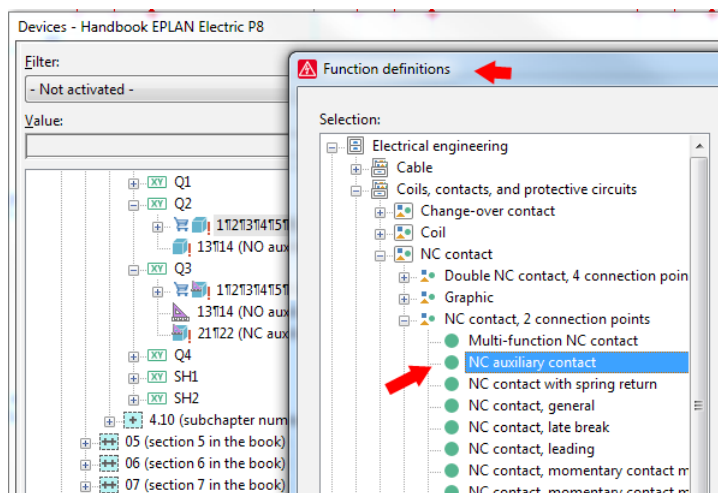


Fig. 5.52 Function definitions dialog

In this dialog, you select the *NC auxiliary contact* function definition and click **OK** to apply it. In the subsequent **PROPERTIES (COMPONENTS)** dialog, EPLAN asks for the *Connection point designation* or the *Device tag*, which is already set by default to the DT selected in the device navigator.

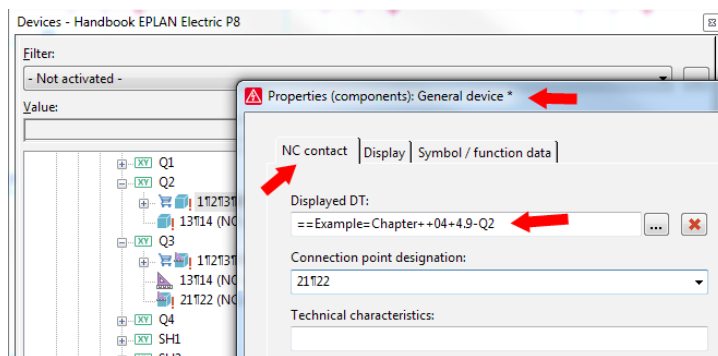


Fig. 5.53 Properties (components) dialog

When you click **OK** to exit the dialog, the *NC auxiliary contact* is assigned to the selected motor overload switch in the device navigator.

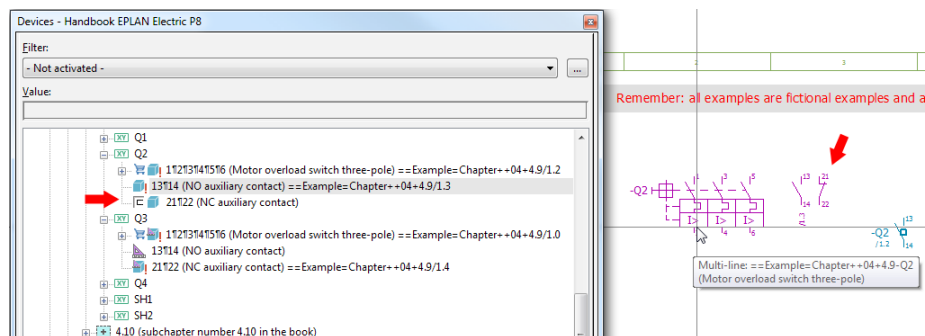


Fig. 5.54 New function on motor overload switch

5.2.8 New functions

To assign one or several new functions to a device, you must select the corresponding device in the device navigator and then call the **NEW FUNCTIONS...** function via the popup menu.

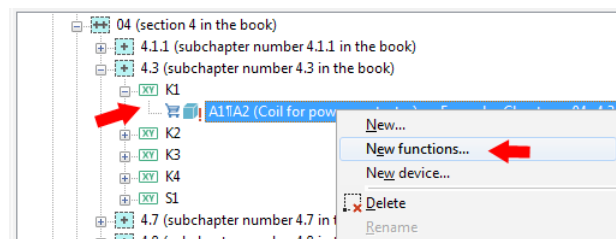


Fig. 5.55 Create new functions

EPLAN starts the **GENERATE FUNCTIONS** dialog and immediately fills the *Full DT* field with the DT selected. Now, using the **MORE** button, the desired function definition is selected from the **FUNCTION DEFINITIONS** dialog and applied to the **GENERATE FUNCTIONS** dialog by clicking **OK**.

Then, a numbering pattern must be defined. If no pattern is defined, EPLAN will not insert any new function (nor can the dialog be exited via the **OK** button). If a pattern is defined, EPLAN will number the new functions automatically when they are added. EPLAN will also count the corresponding number of new functions and transfer this information to the DT as well.

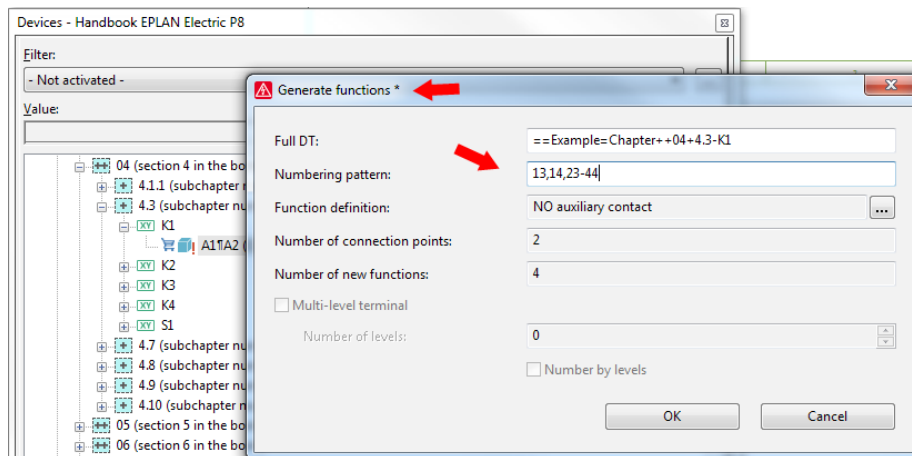


Fig. 5.56 Generate new functions

Once all these entries have been made correctly, you can exit the dialog by clicking OK. EPLAN closes the dialog and transfers the new functions.

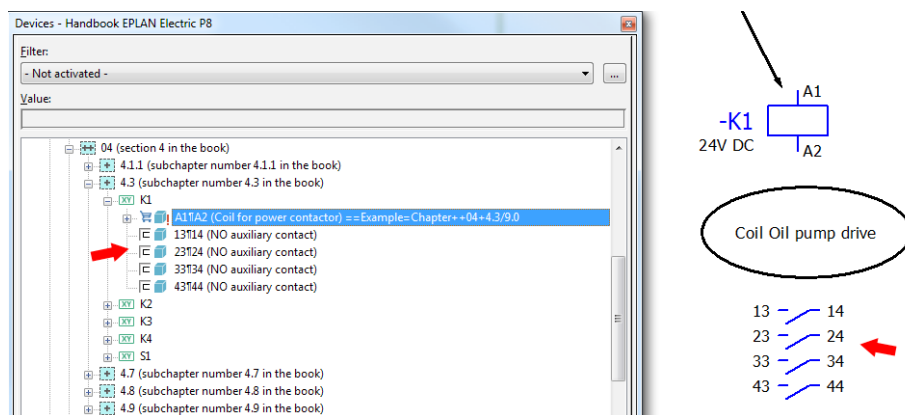


Fig. 5.57 Device with new functions

5.2.9 New device

To insert a new device in the device navigator you select the **NEW DEVICE...** function in the popup menu of the device navigator.

When the **NEW DEVICE...** function is selected, EPLAN starts *parts management*. You select the relevant device and click OK to apply it.



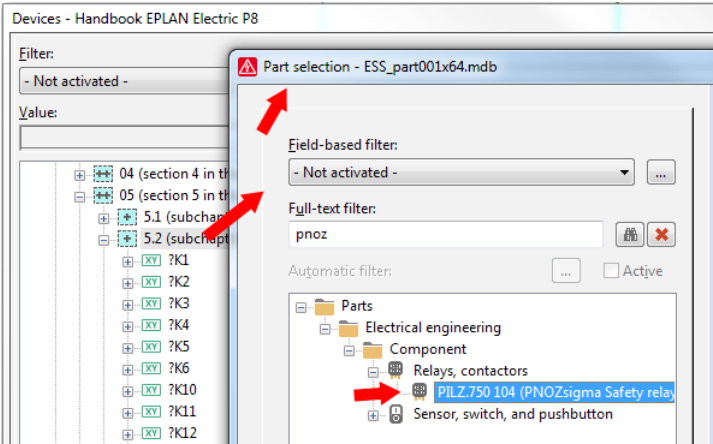


Fig. 5.58 Select device

EPLAN sorts the device according to the *identifier* into the device navigator. It can still be edited. Open the popup menu and select the **PROPERTIES** function, which then opens the **PROPERTIES (COMPONENTS)** dialog.

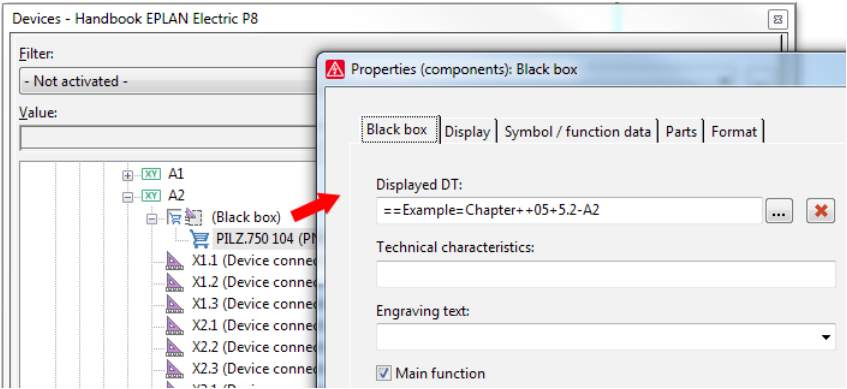
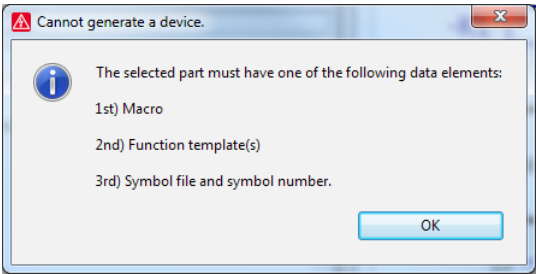


Fig. 5.59 Subsequent device editing



Fig. 5.60
Prompt if a device
cannot be generated

Only new devices with at least one of the following properties, *Macro*, *Function template*, *Symbol file* or *Symbol number*, can be assigned in the device navigator. If this is not the case, then EPLAN displays an appropriate message.



■ 5.3 Terminal strip navigator

There is not much to say about terminal strips. The **terminal strip navigator** helps the user perform many repetitive actions, for example numbering terminals.

Besides the actual terminal strip navigator, the PROJECT DATA/TERMINAL STRIPS menu contains three additional functions. These can be executed directly on the selected elements on a page or on selected pages in the **page navigator**. The **terminal strip navigator** also has a popup menu, accessible by right-clicking the mouse or clicking the  button.

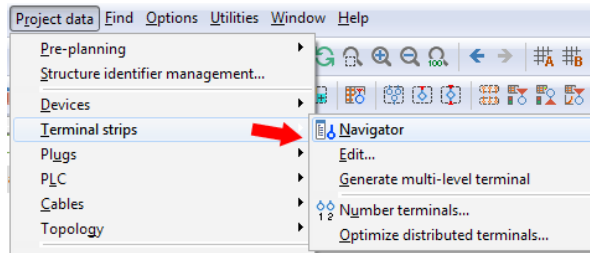


Fig. 5.61
Terminal strip menu

5.3.1 Edit (terminal editor)

5.3.1.1 General information about the terminal editor

The EDIT function in the PROJECT DATA/TERMINAL STRIPS menu opens the EDIT TERMINAL STRIPS dialog. This dialog contains different functions that can be executed or used via the popup menu and/or the upper buttons for editing terminals. Examples of these functions are *Number terminals*, *Move*, *Add strip accessories*, and *delete* or *Generate or ungroup multi-level terminals*.

Like all EPLAN menus, the popup menu in the terminal editor can be called up via the right mouse button.



NOTE: Most functions available in the terminal editor are identical to those that can be called in the terminal strip navigator. Further explanation will therefore not be provided.

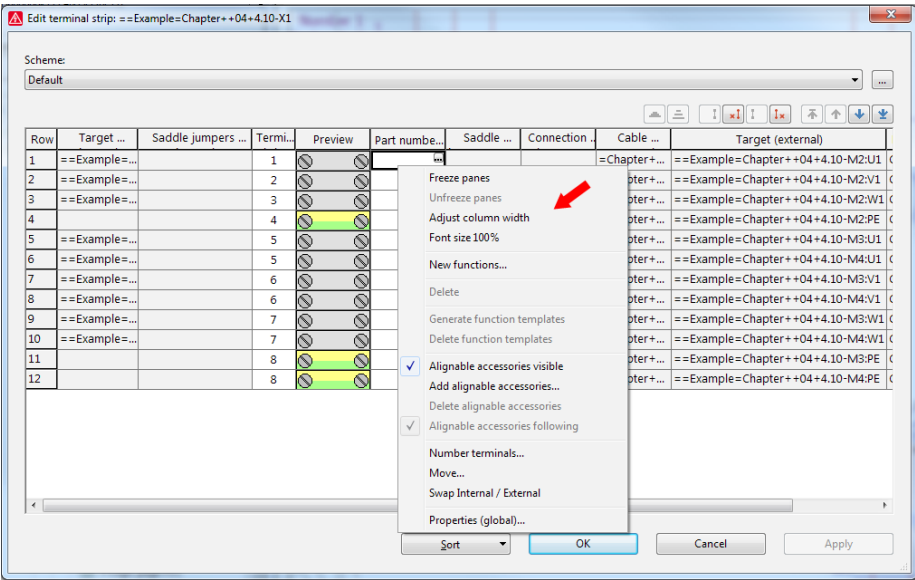
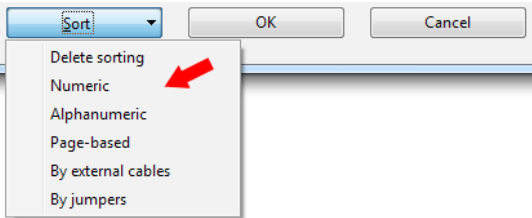


Fig. 5.62 Edit terminal strips dialog

Additional functions for editing terminals are located in the lower area of the dialog after clicking the **Sort** button.

Fig. 5.63 Sort button and its functions



The **Sort** button opens another menu containing functions for sorting terminals. The **DELETE SORTING** function, for example, restores the existing sorting of the terminals back to the default. The **NUMERIC** function sorts the terminals by number, and the **ALPHANUMERIC** function sorts terminals by the alphanumeric components in the terminal designation.

The other buttons, **OK** and **CANCEL**, speak for themselves. The **APPLY** button applies the changes in the **EDIT TERMINAL STRIPS** dialog, but does not close the dialog.



To edit a terminal strip using the terminal editor, you first select the terminal strip in the schematic or in the terminal strip navigator (you can also just select one terminal in the terminal strip to be edited) and then select the **EDIT** function in the **PROJECT DATA/TERMINAL STRIPS** menu.

TIP: To change the row height in the **Edit terminal strips** dialog, position the cursor on a row and then pull the row while holding down the left mouse button to make it smaller or larger.

Row	Target ...	Saddle jumpers ...	Termini...	Preview	Part numbe...	Saddle ...	Connection	Cable ...
1	==Example=C hapter++04...		1					=Chapter+4. 10-W1
2	==Example=C hapter++04...		2					=Chapter+4. 10-W1
3	==Example=C hapter++04...		3					=Chapter+4. 10-W1
4			4					=Chapter+4. 10-W1

Fig. 5.64 Change row height

EPLAN opens the **EDIT TERMINAL STRIPS** dialog. The terminals are sorted here according to the standard. Click the **SORT** button and select the **DELETE SORTING** item. EPLAN sorts the terminal strips as follows: The existing device position and sort code at the terminal are deleted.

**NOTE:** This will also break up defined terminal devices. Thus, terminals are sorted by their designation (1, 4 or 17) and/or based on Sorting (graphical).

Row	Target ...	Saddle jumpers ...	Termini...	Preview	Part numbe...	Saddle ...	Co
1	==Example=...		1				
2	==Example=...		2				
3	==Example=...		3				
4			4				
5	==Example=...		5				
6	==Example=...		5				
7	==Example=...		6				
8	==Example=...		6				
9	==Example=...		7				

Fig. 5.65 Newly sorted terminal strip

For example, to change the sequence of terminals 1, 2 and 3 (from 1-2-3 to 3-2-1), you select each of the terminals and use the buttons or the associated keyboard shortcuts to move them to the desired position.

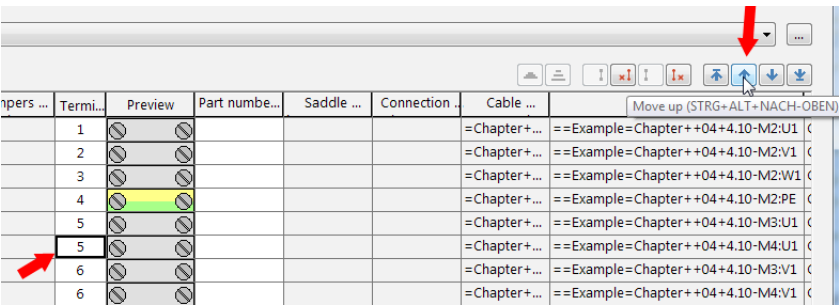


Fig. 5.66 Manually moving a terminal

Fig. 5.67
Renaming terminal
designation

To give one or more terminals another terminal designation, you need only edit the **terminal designation** field directly. EPLAN writes the changed terminal designation to the corresponding terminals immediately after you save the changes and exit the dialog.

Row	Target ...	Saddle jumpers ...	Termi...	Preview
1	==Example=...		1	
2	==Example=...		5	
3	==Example=...		2	
4	==Example=...		3	
5			4	

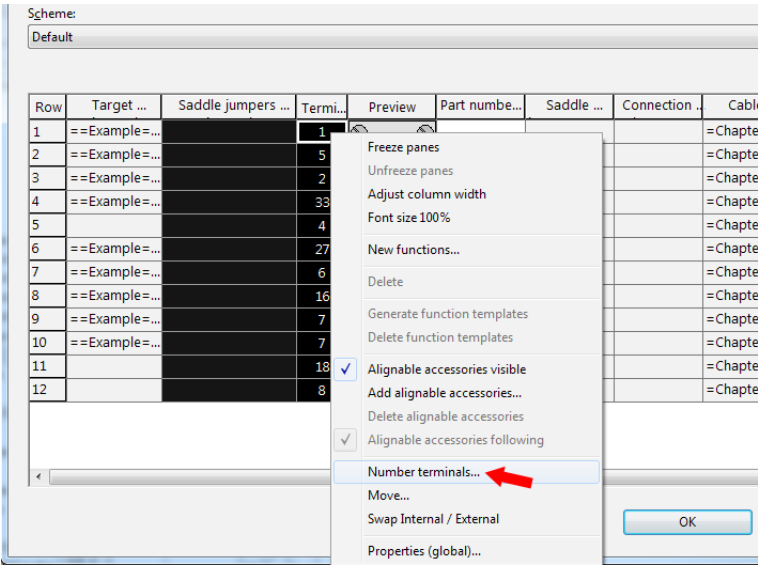


Fig. 5.68 Numbering terminals

Terminals can also be numbered in the dialog. Simply select the desired terminals with the mouse, open the popup menu and select the **NUMBER TERMINALS** menu item. EPLAN then starts the **NUMBER TERMINALS** dialog where the settings are to be made. EPLAN can then number the terminals.

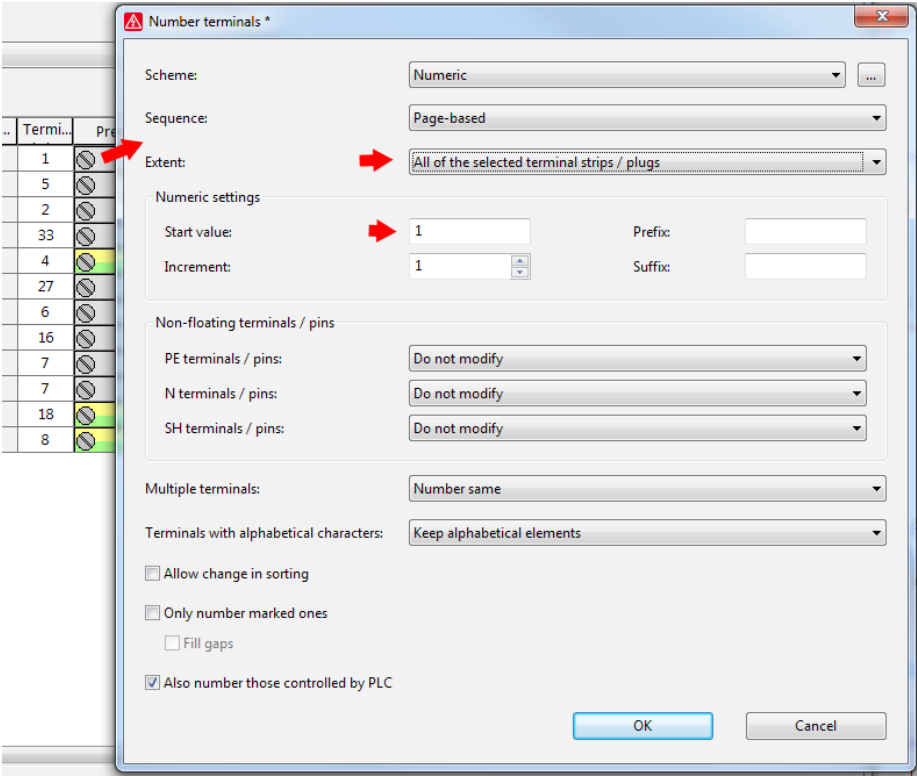


Fig. 5.69 Number terminals dialog

The newly numbered terminal strip will then look like the one in Fig. 5.70.

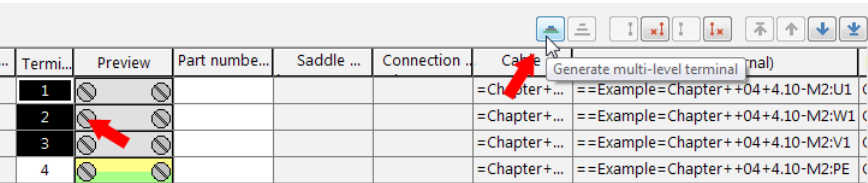
Row	Target ...	Saddle jumpers ...	Termi...	Preview	Part numbe...	Saddle jum...
1	==Example=...		1			
2	==Example=...		2			
3	==Example=...		3			
4			4			
5	==Example=...		5			
6	==Example=...		6			
7	==Example=...		7			
8	==Example=...		8			
9	==Example=...		9			
10			10			
11	==Example=...		11			
12			12			

Fig. 5.70 Numbered terminal strip

5.3.2 Generate multi-level terminal

Multi-level terminals can also be generated conveniently in the terminal editor. It is quite easy to do. Select the terminals to be used for a multi-level terminal.

Fig. 5.71
Selecting terminals



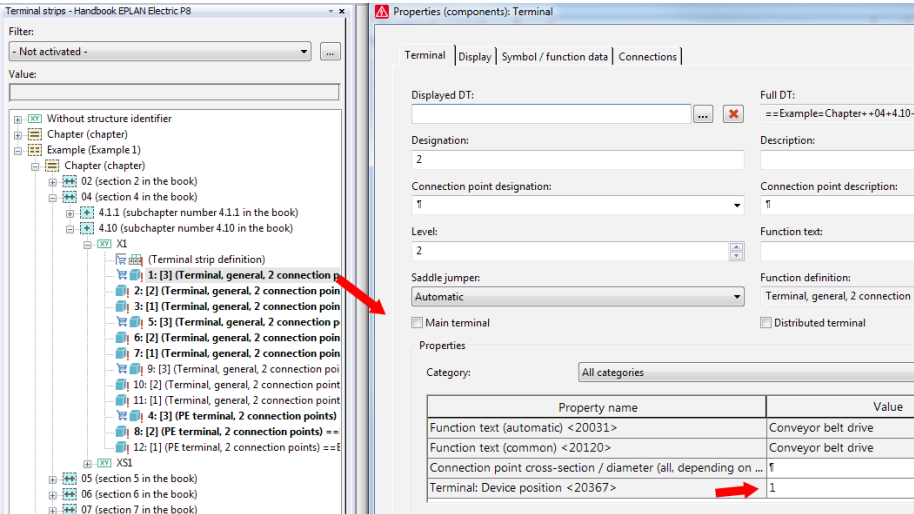
Then, click the  button (GENERATE MULTI-LEVEL TERMINAL). EPLAN groups the selected terminals to form a multi-level terminal.

Fig. 5.72
Generated multi-level terminals

Row	Target ...	Saddle jumpers ...	Termi...	Preview	Part numbe...	Saddle ...	Connection ..
1, 2, 3	==Example=...		1				
	==Example=...		2				
	==Example=...		3				
4, 5, 6	==Example=...		5				
	==Example=...		6				
	==Example=...		7				
7, 8, 9	==Example=...		9				
	==Example=...		10				
	==Example=...		11				
10, 11, 12			4				
			8				
			12				

During this action, EPLAN allocates the levels and assigns a device position each to these multi-level terminals (Property **Terminal: Device position** <20367>).

Fig. 5.73
Property Terminal:
Device position



There are also other options in the terminal editor, but they are self-explanatory:

- Generate saddle-jumpers for selected terminals (externally or internally)
- Delete saddle-jumpers for selected terminals (externally or internally)
- Individual part selection per terminal
- Swap internal/external connection points for selected terminals

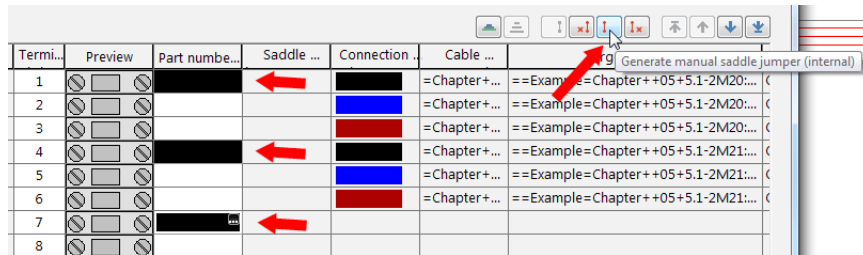


Fig. 5.74 Other options in the terminal editor

5.3.3 Number terminals

As a result of various project actions in EPLAN, such as copying or copying of pages from other projects, terminals can end up with the wrong or completely incorrect designations. To avoid having to manually renumber all these terminals, the **PROJECT DATA/TERMINAL STRIPS** menu offers the **NUMBER TERMINALS** function.

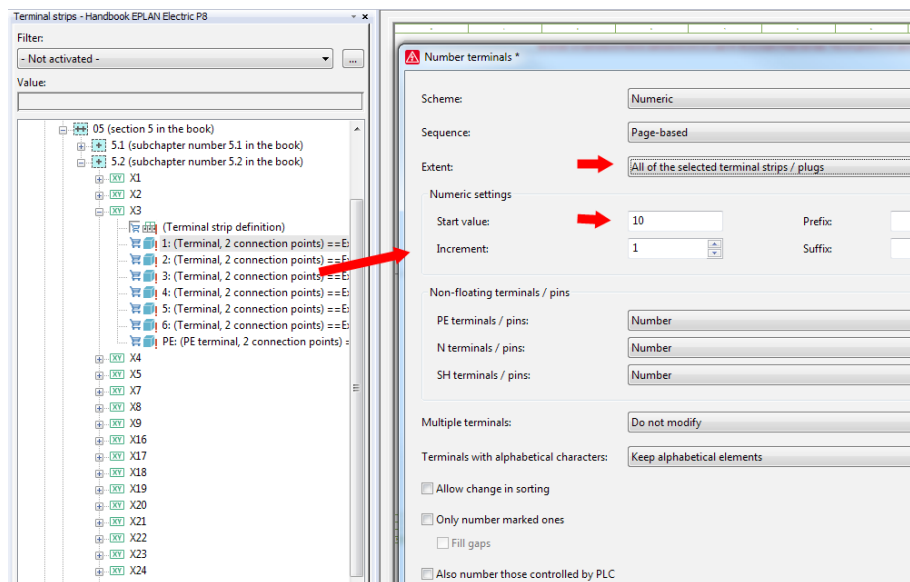


Fig. 5.75 Number terminals dialog

You can number terminals using more than a simple scheme such as numbering integer values from 1 to 10. EPLAN also allows you to define your own schemes, e.g. for initiator terminals.



The terminal strip to be renumbered is selected in the schematic (or in the terminal strip navigator). You only have to select one (any) terminal belonging to this terminal strip.

You then start the **NUMBER TERMINALS** dialog via the **NUMBER TERMINALS** item in the **PROJECT DATA/TERMINAL STRIPS** menu.

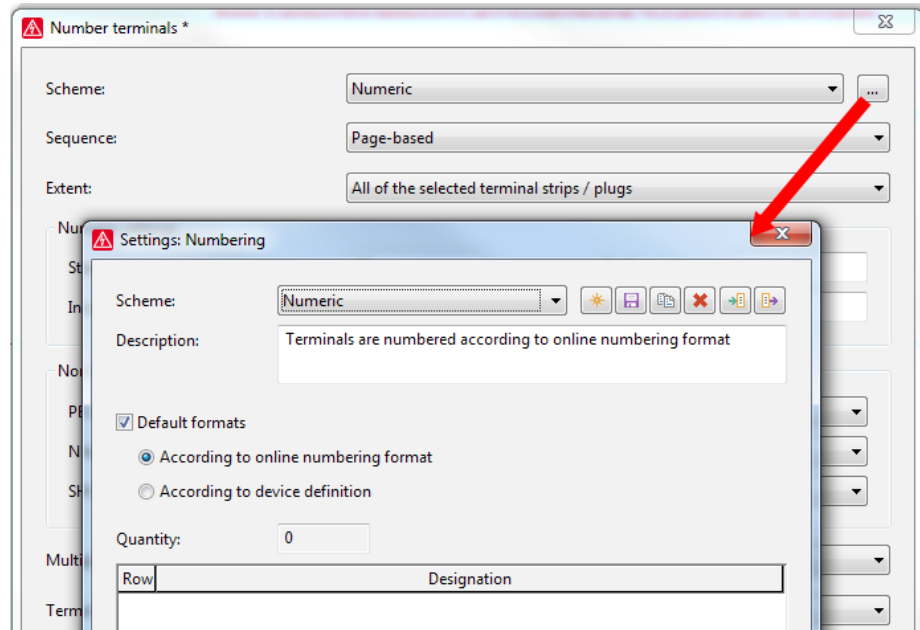


Fig. 5.76 Number terminals dialog with selection of numbering scheme

In the **NUMBER TERMINALS** dialog, the desired scheme is now selected in the **Scheme** selection field; in the example, this is the *Numeric* scheme. The scheme can also be changed before numbering by using the  button or by generating a new scheme in the **SETTINGS: NUMBERING** dialog.

Before you start the actual numbering, you should check and adjust the other settings, such as the numeric settings, the extent, and the handling of PE terminals.

When you click **OK** in the **NUMBER TERMINALS** dialog is clicked, EPLAN numbers the terminal strips as specified.

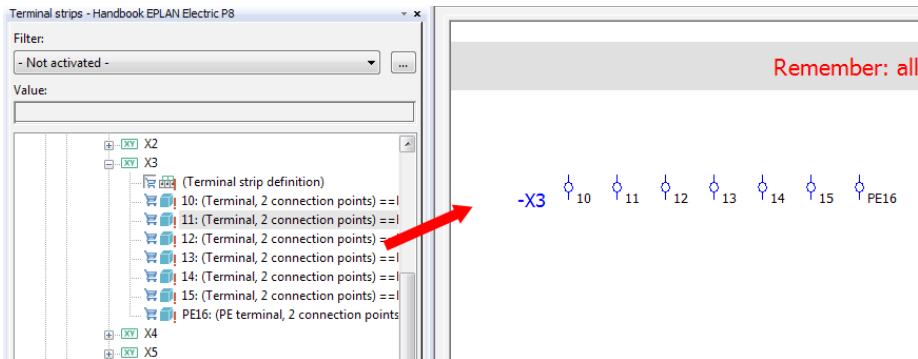


Fig. 5.77 Newly numbered terminals

This simple example is only meant to illustrate the process of using the **NUMBER TERMINAL STRIPS** function. More complex numbering tasks can be accomplished by making use of the many possible settings in terminal schemes.

5.3.3.1 Number terminals settings

In the **NUMBER TERMINALS** dialog, in addition to the basic numbering settings, there are other options, which are explained below.

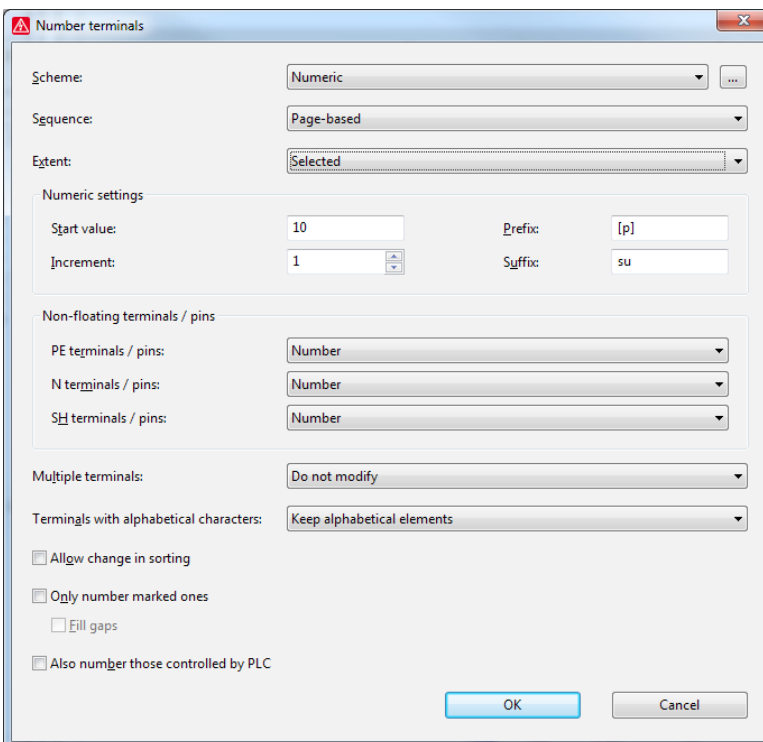


Fig. 5.78
Number terminals
dialog

Scheme: The desired scheme can be set here from a selection list. You can add as many schemes as you want to this selection list.

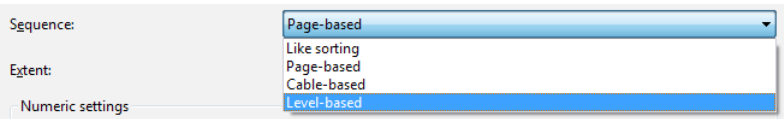


Fig. 5.79 Sequence setting

Sequence: The numbering sequence is set in this selection list. Should the sequence be oriented to pages, cables or levels, or should the existing sorting method be applied?

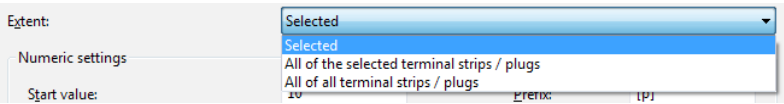


Fig. 5.80 Extent setting

Extent: This setting determines the range of the terminals to be numbered. What should be numbered – only selected terminals, the entire terminal strip, or all terminals in the project?

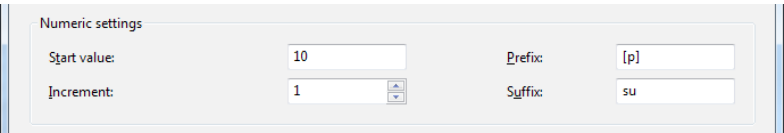


Fig. 5.81 Numeric settings

Numeric settings: This is where the start value, increment, prefix ([n] character preceding the terminal designation) and suffix ([n] character after the terminal designation) are set.

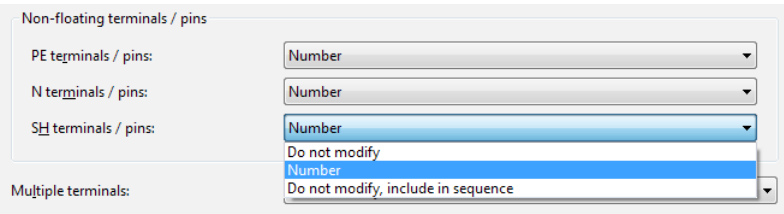


Fig. 5.82 Settings for specific terminal types

Non-floating terminals/pins: This is where settings for PE, N and SH terminals are set. Should they stay in the set sequence and not be changed, numbered or modified? The potential type of the terminals and associated function definition are crucial here!

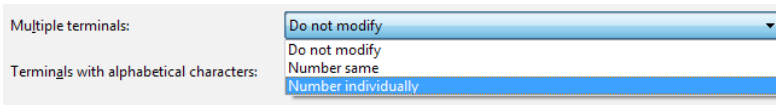


Fig. 5.83 Settings for multiple terminals

Multiple terminals: How should multiple terminals be handled during numbering operations? Should they be left unchanged, numbered all the same, or numbered individually?

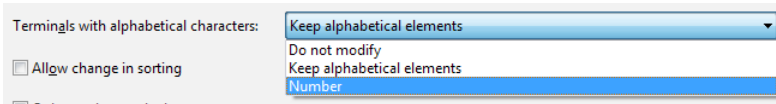


Fig. 5.84 Settings for alphabetically designated terminals

Alphabetically designated: This setting means that terminal designations that contain alphabetical components are taken into account. This setting determines whether these will be left unchanged and the components retained, or whether these will be completely renumbered. In the latter case, the alphabetical components would be lost.

In addition to these settings, there are four more options.

Allow change in sorting: This affects the terminal storing on the terminal strip. An active option re-sets the (empty) sort codes. But manual sort codes are retained. A non-active option leaves the sorting as it was previously assigned.

Only number marked ones: Only those terminals with the ? placeholder are numbered. Depending on the *Fill gaps* setting, gaps between the terminal numbers (where they already exist, they will continue to be applied) are filled, or, if this option is not used, the terminals are completely renumbered.

The last setting, *Also number those controlled by PLC*, only numbers those terminals that are controlled by PLC.

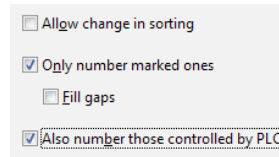


Fig. 5.85
Other options

5.3.4 New

The **NEW...** function is basically the same as in the **device navigator**. In contrast to the device navigator, which can insert any function definition, only terminal or terminal strip function definitions can be inserted in the **terminal strip navigator**. Only one new terminal is inserted each time!

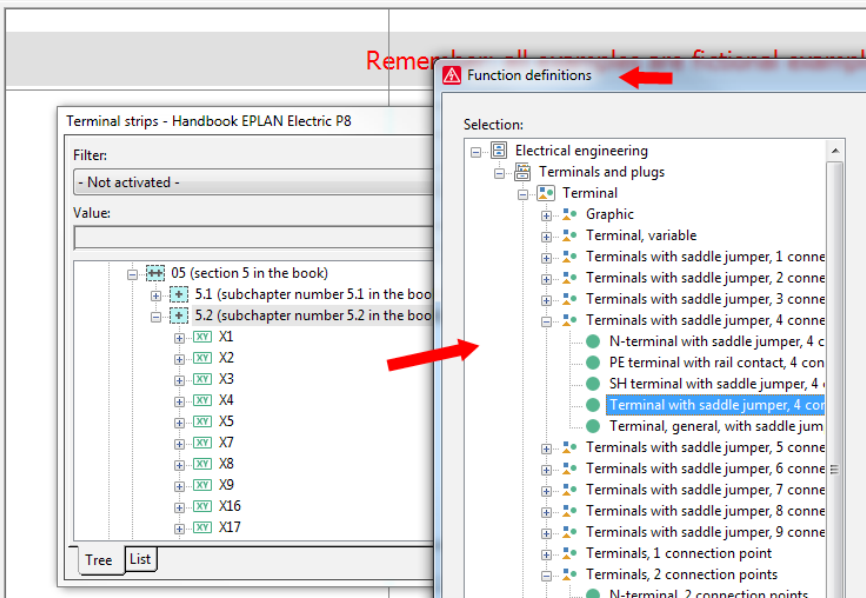


Fig. 5.86
Inserting new terminals

5.3.5 New functions

The **NEW FUNCTIONS...** function in the **terminal navigator** allows fast generation of complete terminal strips using specific (own) numbering schemes.



The **NEW FUNCTIONS...** function is called up in the **terminal strip navigator** popup menu. EPLAN opens the **GENERATE FUNCTIONS** dialog, which is initially empty, or a terminal strip designation is applied. The full DT is already displayed, depending on the current focus in the terminal strip navigator.

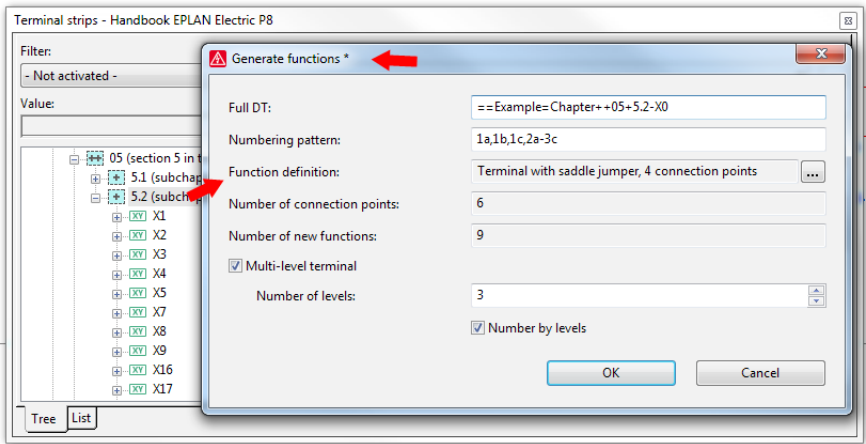


Fig. 5.87
Generate functions
dialog

Here you should select/enter the full DT (the terminal strip designation), the numbering pattern and the setting defining whether or not these are multi-level terminals.

It is also possible to select a different function definition instead of the default terminal. Just click the  button. EPLAN switches to the **FUNCTION DEFINITIONS** dialog where you can select a different terminal and apply it by clicking the OK button.

Generate functions

However, this is not a precondition. All function definitions can be easily changed later using block editing in the terminal strip navigator.

We return once more to generation of the functions: The functions can only be generated when all entered data is correct (EPLAN will enable the OK button). If an error exists in the numbering pattern, e.g. a missing comma, then the OK button is not enabled. You must check for this. When the OK button is clicked, EPLAN generates a terminal strip according to the specified options.

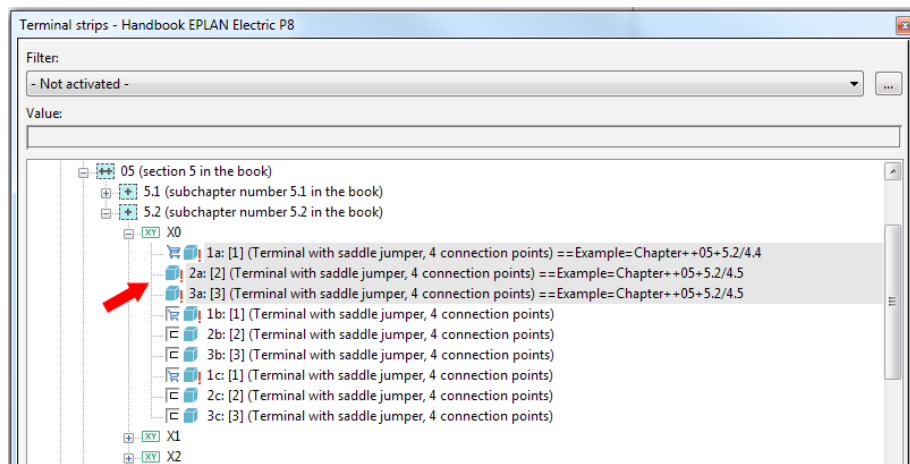


Fig. 5.88 New terminal strip with placed and unplaced terminals



NOTE: The newly created terminals do not have any parts data. This data must (can) be assigned later using device selection.

5.3.6 New terminals (devices)

In addition to the easy generation of new terminal strips and new terminals, it is also possible in the terminal strip navigator popup menu to generate new terminals as devices on the basis of existing parts and their function templates.

Start the **NEW TERMINALS (DEVICES)...** menu entry from the popup menu.

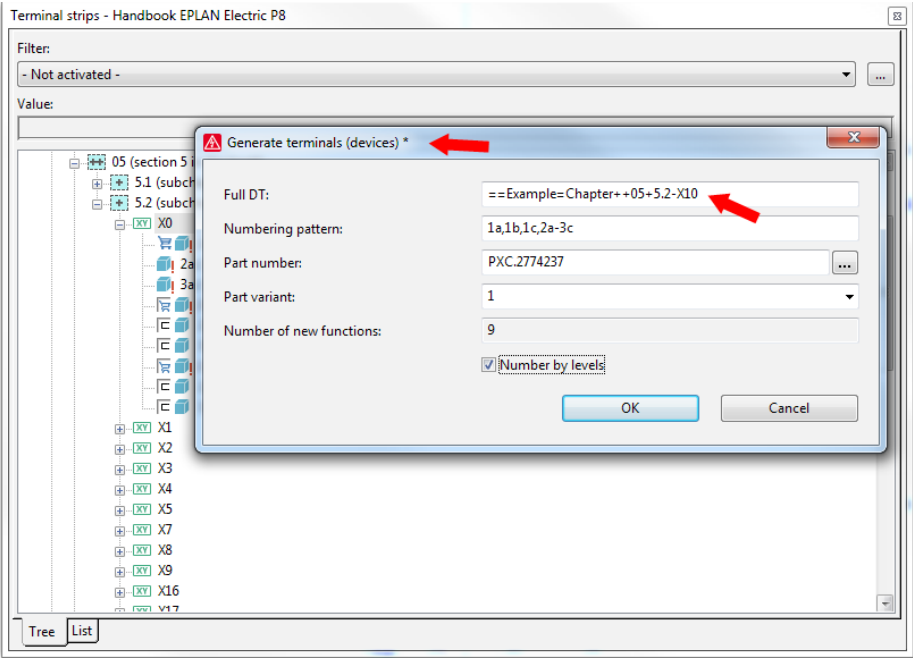


Fig. 5.89 Generate terminals (devices) dialog



EPLAN opens the **GENERATE TERMINALS (DEVICES)** dialog. The entries are similar to the previous section. You need to select a DT, a numbering pattern, and, most importantly, a part (the device) from parts management.

When all entries are correct, you can start device generation by clicking the **OK** button. EPLAN saves the functions in the navigator as unplaced functions.

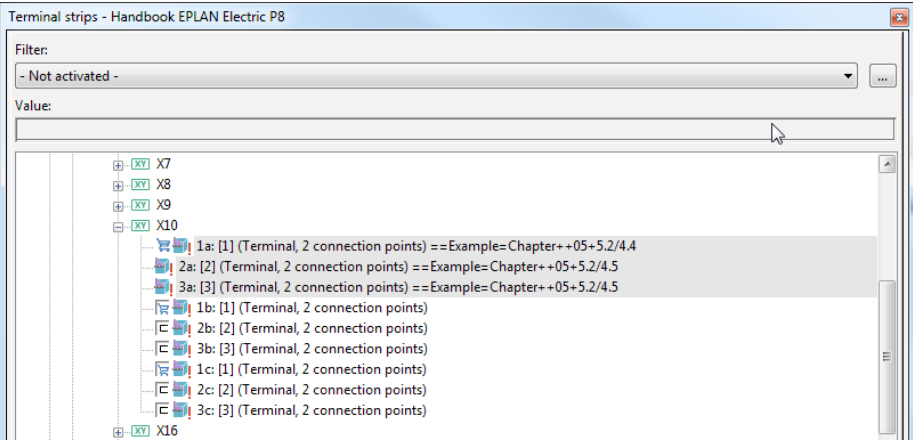


Fig. 5.90 Generated, new terminals (devices)

In contrast to the last section, these new terminals are real devices. They already have parts data.

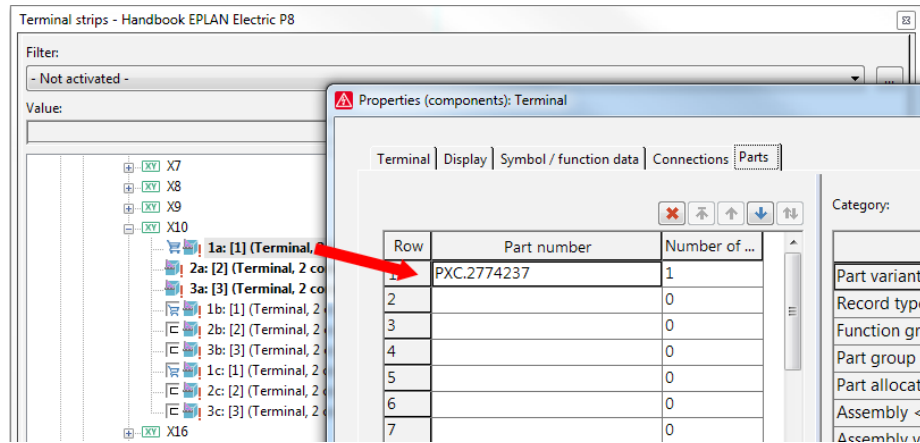


Fig. 5.91 Terminal parts data

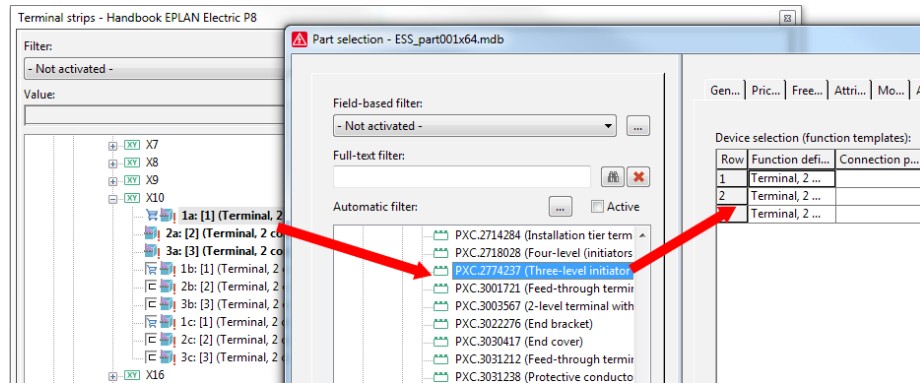


Fig. 5.92 Function template of the terminal part

5.3.6.1 Main terminals/Auxiliary terminals

It is important to know that, in addition to the generation of terminals or of terminals as devices, EPLAN has introduced new terminal functions and a new term in Version 2.0 – the **Main terminal**.

What is a main terminal? It is comparable to a main function; a terminal with multiple function definitions can only be identified once as a main terminal. All of this terminal's other functions cannot be designated as main functions. These are so-called **auxiliary terminals**.

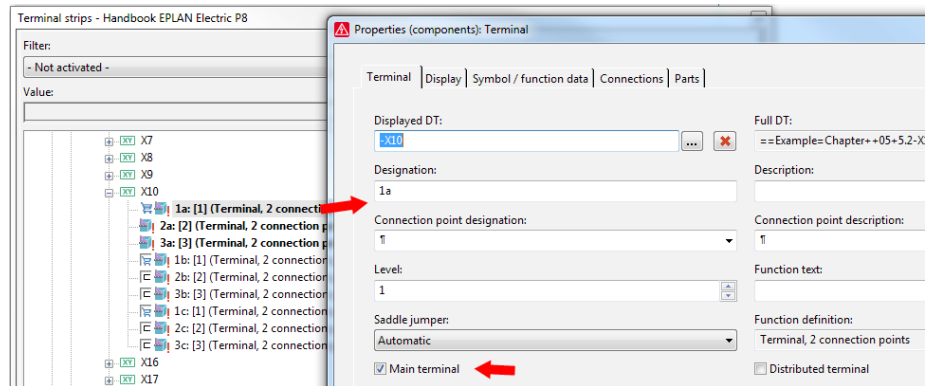


Fig. 5.93 Main terminal

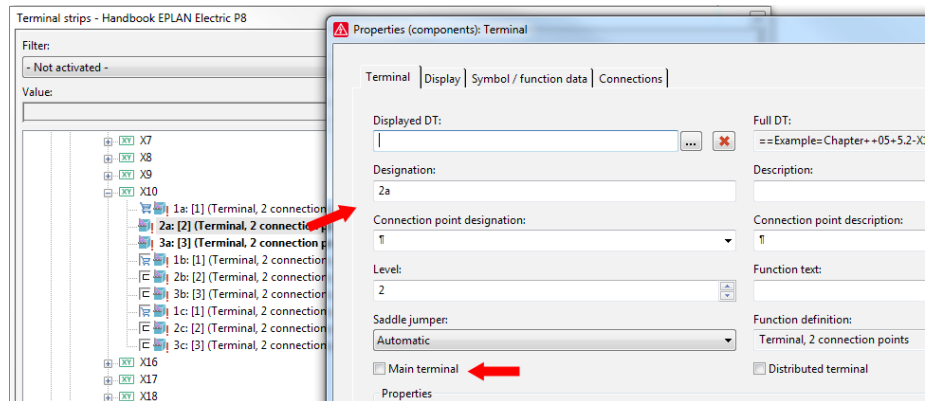


Fig. 5.94 Auxiliary terminal

Only main terminals can have *parts data*. These new main and auxiliary terminals are now handled like all other devices and can be used like other devices in EPLAN. You therefore have the option of using or placing some of this terminal's functions, e.g. as a grounding contact or a coil. Main and auxiliary terminals have the same relationship to each other. Device selection can be performed for these main terminals. The terminal strip definition, however, remains a terminal strip's **main function**.

5.3.7 View

The terminal strip navigator provides two different views of its devices (here: terminals and terminal strips). You switch to this view via the popup menu of the right mouse button.

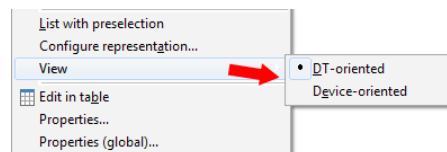


Fig. 5.95
Selection of view

5.3.7.1 DT-oriented

The DT-oriented view is the familiar view of the DT of a terminal strip.

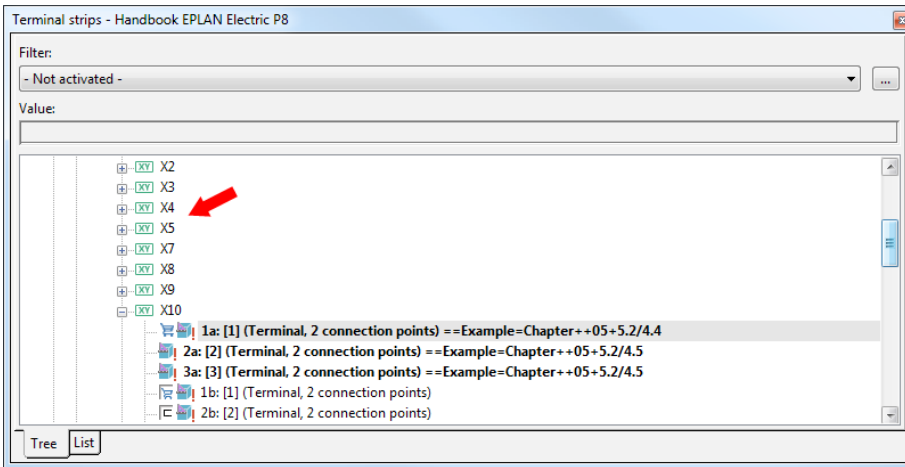


Fig. 5.96 DT-oriented view

5.3.7.2 Device-oriented

The second, device-oriented view switches the view to the devices, such as multi-level terminals.

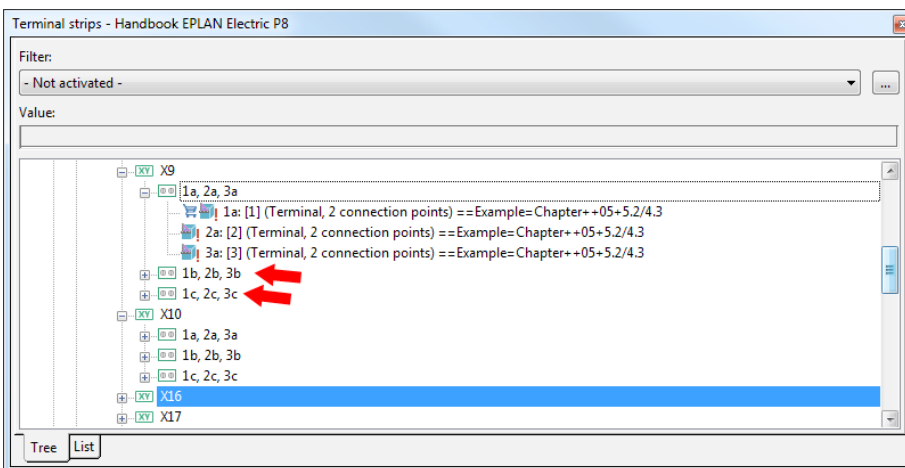


Fig. 5.97 Device-oriented view

■ 5.4 Plug (female pin) navigator

In addition to terminals, there are also plugs and sockets. In principle, the **plug navigator** offers the same functions as those used for editing terminals and terminal strips. This section will therefore only cover the plug/socket functions that are different from the functions provided in the terminal strip navigator.

Plug is a generic term in EPLAN. Plugs are distinguished by male and female pins. Corresponding function definitions exist for these, and the various menu items are also labeled as such.

5.4.1 Edit (plug/female pin editor)

In addition to the actual **plug navigator** (which also includes the female pins), **plug** project data also has a submenu with additional special functions. The **EDIT PLUG** function works in a way very similar to the **EDIT TERMINAL** function. In this case, you edit only male pins or female pins.

After you call up the menu, EPLAN displays the **EDIT PLUG** dialog. The functions available in the popup menu, such as *Number pins* or *Edit properties*, can be used here.

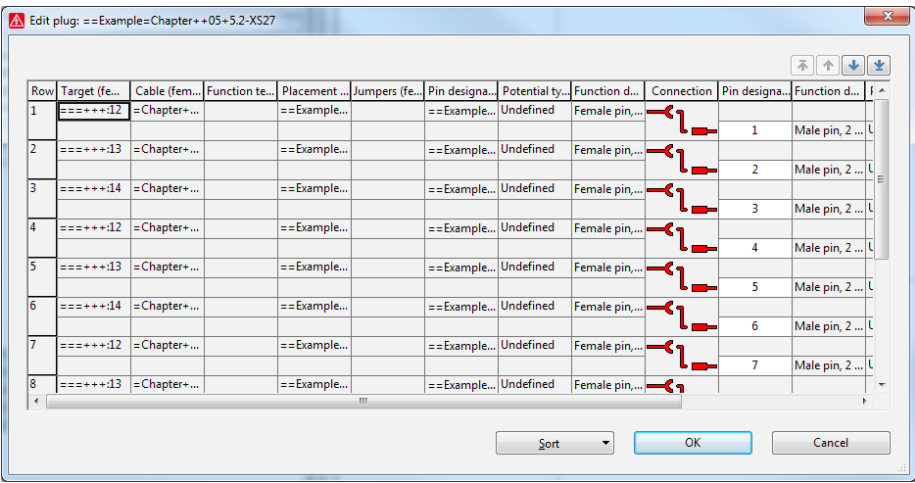


Fig. 5.98 Edit plug dialog

5.4.2 Number pins

This function is identical to the **NUMBER TERMINALS FUNCTION** – with one exception. In the **NUMBER PINS** dialog, the **MULTIPLE TERMINALS** option is not possible for plugs, of course.

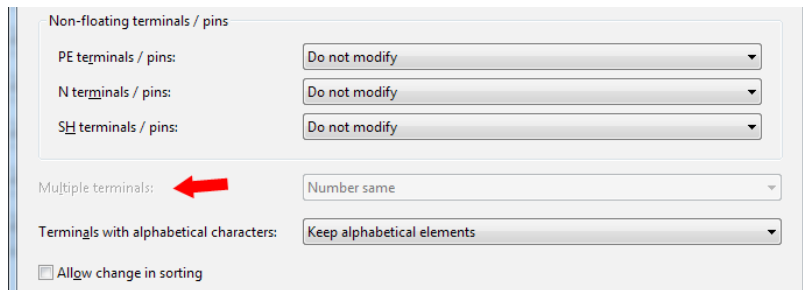


Fig. 5.99 Number plugs dialog

5.4.3 New

The popup menu of the **plug navigator** (accessible by right-clicking) contains the **NEW ...** menu item. The procedure here is identical to the **NEW...** function in the **terminal navigator**. Here, too, a new function definition can be selected. In contrast to terminals, this is only possible for plugs and sockets.

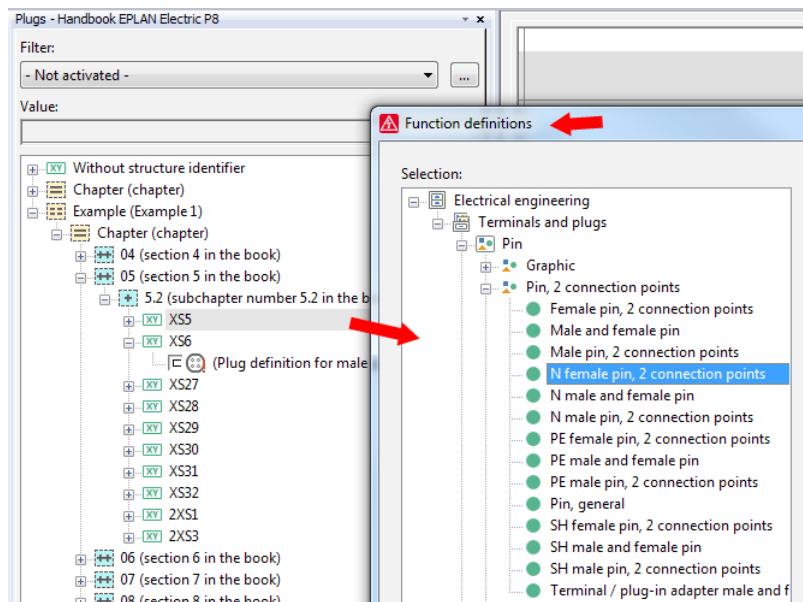


Fig. 5.100 Creating a new, individual plug or female pin

5.4.4 New functions

New plug strips are created using the popup menu item in the **plug navigator** in exactly the same way as in the **terminal strip navigator**. Whereas the terminal strip navigator creates new terminal strips, in the plug navigator you create new plugs, sockets, and plug/socket strips – depending on the function definition settings.

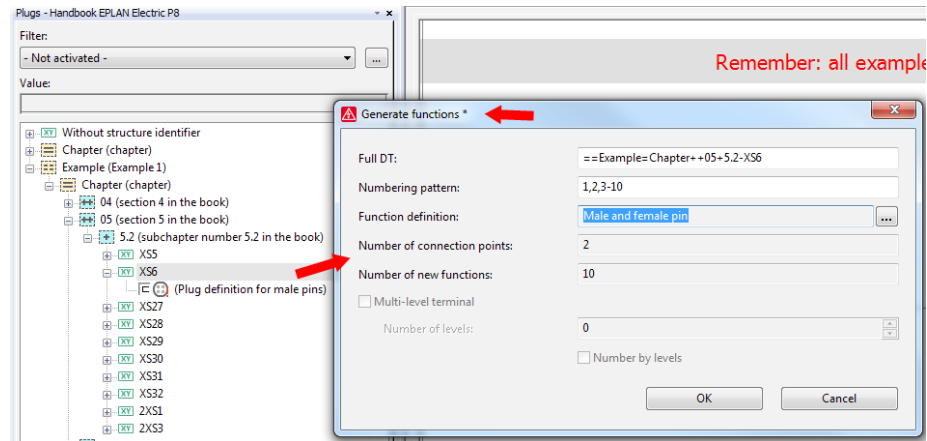


Fig. 5.101 Example of generating new functions

In contrast to generating new terminal functions, here it is not possible either to create multi-level terminals. The remaining procedures are the same as in the terminal strip navigator.

5.4.5 Generate plug definition (plug and/or female pin)

As in the **terminal strip navigator**, you select new plugs in the **plug navigator** via the popup menu's **GENERATE PLUG DEFINITION**. Then you select **ONLY MALE PINS** or **ONLY FEMALE PINS** in the **MALE AND FEMALE PINS** field.

If, for example, *Only male pins* is selected, EPLAN displays the **PROPERTIES (COMPONENTS)/PLUG DEFINITION** dialog and enters the correct function definition. The remaining entries, such as device tag or form, can be entered and updated accordingly.

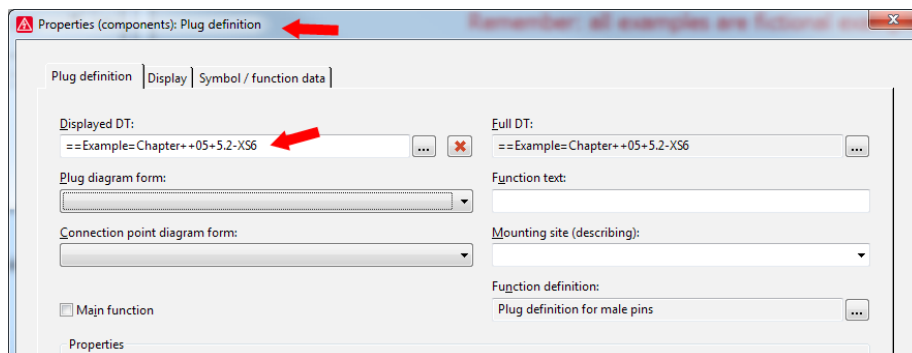


Fig. 5.102 New definition for male pins

5.4.6 Generate pin (plug and/or female pin)

The GENERATE PIN function generates a new pin for the plug, socket or plug strip selected in the plug navigator.

You select the GENERATE PIN function from the popup menu in the plug navigator and then select, as in this example, the MALE PIN entry. EPLAN immediately opens the PROPERTIES (COMPONENTS)/PIN dialog. The device tag of the plug is automatically applied.

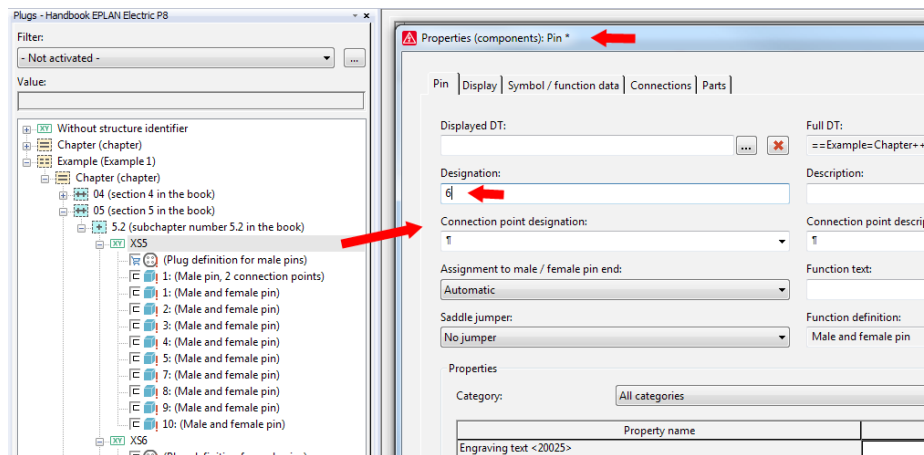


Fig. 5.103 Creating a male pin

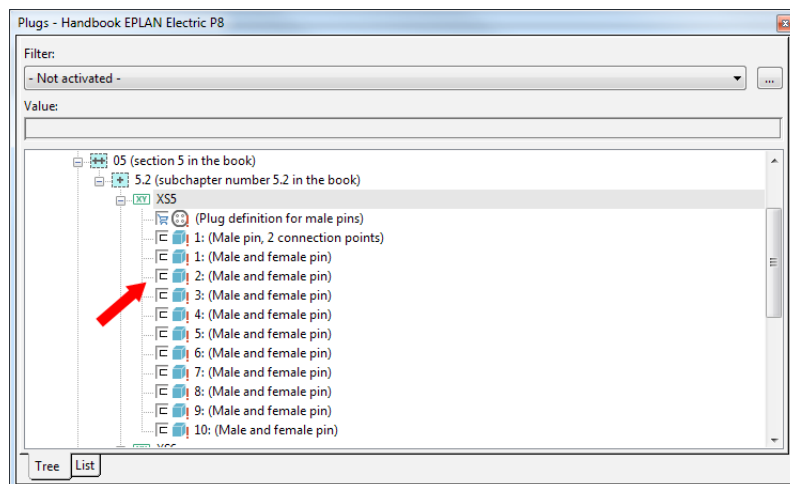


Fig. 5104 Unplaced male pins

In the **Displayed DT** field, use  to BRANCH to the DT SELECTION dialog. All the plug's pins are listed here. A male pin can be designated and then assigned to the plug as an unplaced male pin by clicking OK.

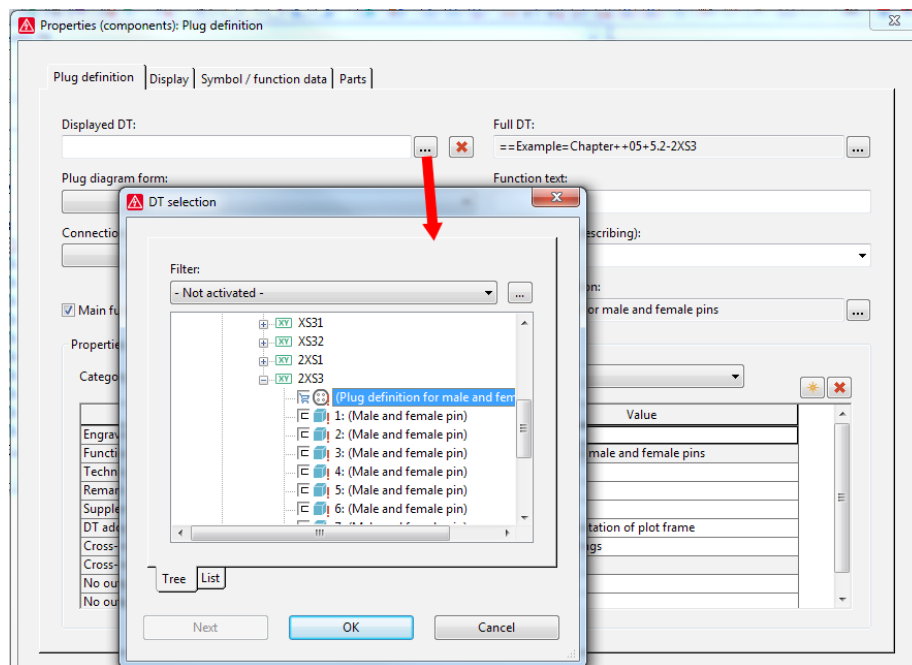


Fig. 5.105 DT selection

■ 5.5 Cable navigator

Cables and their conductors connect the individual devices. The **cable navigator** provides an overview of all used and unplaced cables in the project.



NOTE: To edit cables with all the functions provided by the cable navigator, such as automatic cable selection, the logical connection data must always be complete and up to date.

In most cases, EPLAN automatically performs **UPDATE CONNECTIONS** before such actions. But in the event that the connection data is not up to date, it is possible that subsequent actions will not work as desired or expected. This is why you should check the *status of connections* (up to date or not) in the status bar (symbol # or *), and update the connections yourself as required.

As I have mentioned before in this book, my personal recommendation is to assign the **UPDATE CONNECTIONS** function to the F11 key. Then updating connections is only one keystroke away.

As with all the other navigators, the cable navigator also has a popup menu with additional functions for cable editing. As usual in EPLAN, the popup menu is called up via the right mouse button or the  button.

5.5.1 Edit

The **PROJECT DATA/CABLES** menu has an **EDIT** menu item. This function allows you to manually edit the assignments of the individual connections (which cable conductor – source/target) without requiring you to manually change the individual conductors of the cable in the schematic.

After various copy and delete actions, the assignment of conductors/terminals for a cable no longer reflects the desired assignments. To conveniently change the assignment, in the schematic you first select the cable to be changed (click the cable symbol) and then open the **EDIT** function in the **PROJECT DATA/CABLES** menu.

EPLAN opens the **EDIT CABLE** dialog. The left field lists the cable with its properties (stored **function templates** – conductor/wire). The cable's determined connections are shown in the right field.



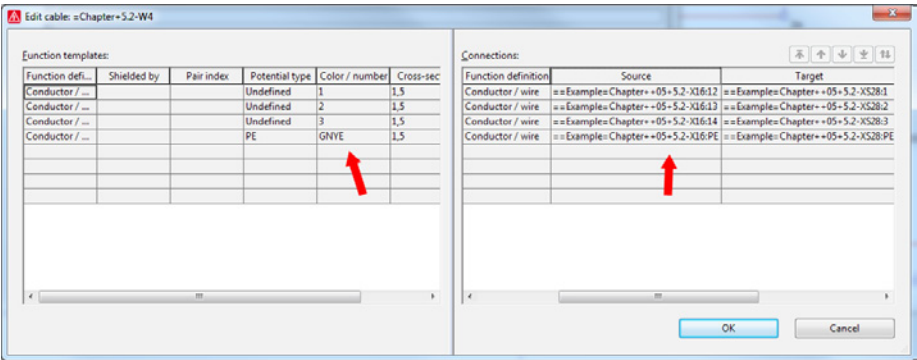


Fig. 5.106 Assignment conductor to source and target DT

The corresponding conductors can now be clicked in the **Connections** field and can then be moved using, for example, the   buttons (move up or down) and assigned to the correct (desired) connections. You can also exchange connection places. Select the two connections that are to be exchanged and click the  button. EPLAN then exchanges the assignments in the table.

The other “wrong” assignments can be corrected in the same way. After you finish changing the assignment, you exit the **EDIT CABLE** dialog. EPLAN then automatically updates the *conductor – source/target* assignments in the schematic, as can be seen in Fig. 5.107 and Fig. 5.108.

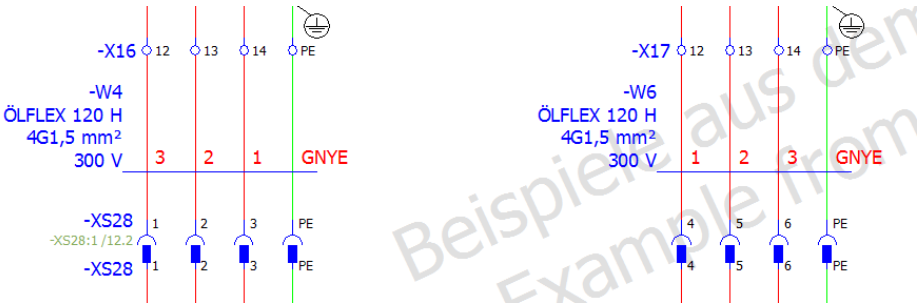


Fig. 5.107 Conductor assignment before the Edit cable dialog

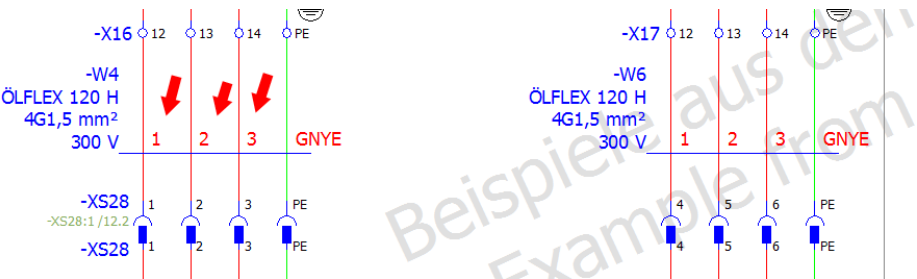


Fig. 5.108 Conductor assignment after exiting the Edit cable dialog

5.5.2 Number (cable DT)

Schematics can be composed of many different macros or other project pages. The **cable DTs** can be very different. The **NUMBER** function in the **PROJECT DATA/CABLES** menu allows these to be consistently numbered across the entire project.

In the schematic or in the **cable navigator**, select the cables to be numbered.

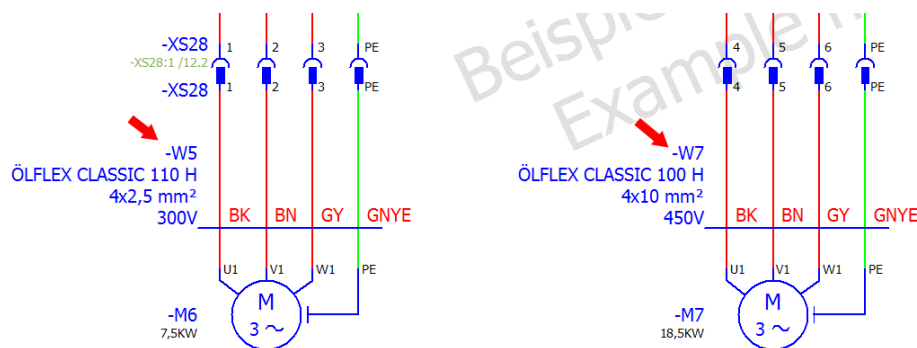


Fig. 5.109 Selected cables

Then call up the **NUMBER** function in the **PROJECT DATA/CABLES** menu. EPLAN opens the **NUMBER CABLES** dialog. The numbering options can be set in the *Settings* selection field.

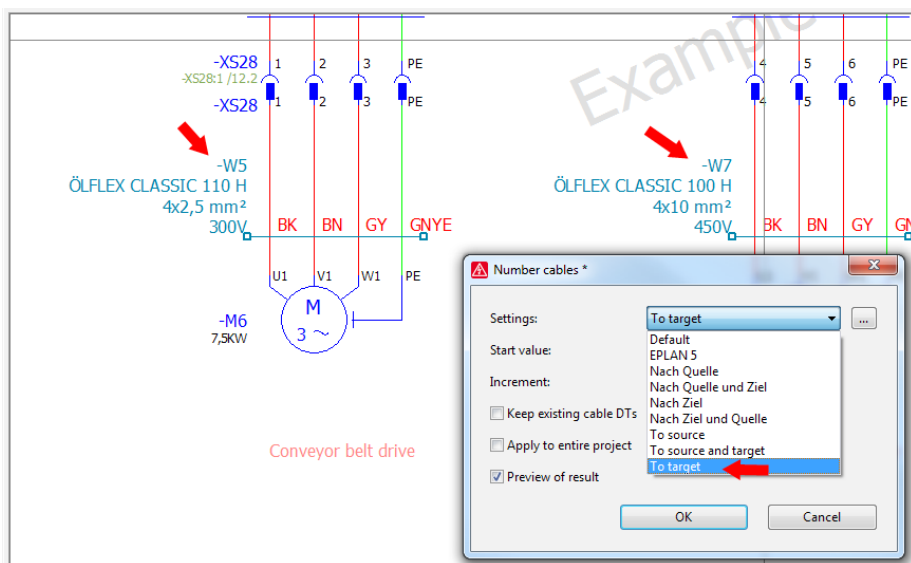


Fig. 5.110 Number cables dialog

Use the  button to switch to the **SETTINGS/CABLE NUMBERING** dialog and create your own scheme for cable numbering, or modify and save an existing scheme.

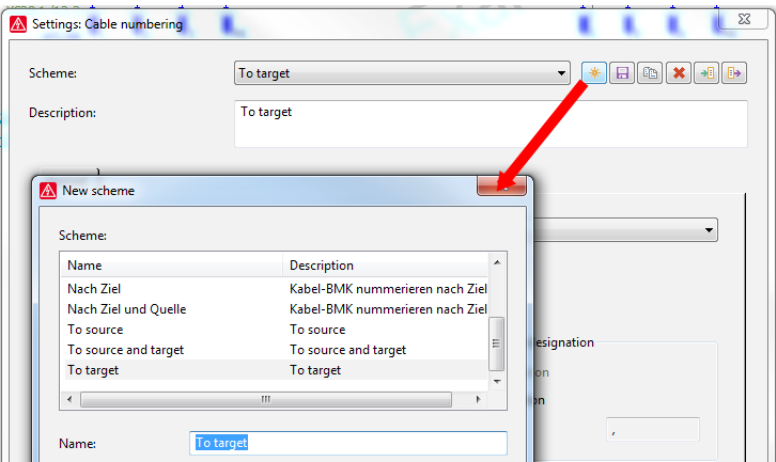


Fig. 5.111 Settings/Cable numbering dialog

In the example, the existing *Scheme: according to target* was selected in the **EDIT CABLE** dialog for numbering the selected cables. Here too it is a good idea to activate the *Preview of result* parameter. This allows the new cable DTs to be checked before they are written back to the schematic, and if necessary, the numbering can be stopped via the **CANCEL** button.

After you click **OK**, EPLAN rennumbers the **cable DTs** according to the selected scheme and displays the result in the **NUMBER CABLES: PREVIEW OF RESULT** dialog.

Number cables: Preview of result		
Row	DT (identifying)	DT (identifying) New
1	=Chapter+5.2-W5	=Chapter+5.2-W(==Example++05-M6)
2	=Chapter+5.2-W7	=Chapter+5.2-W(==Example++05-M7)

Fig. 5.112 Preview of result dialog

If the result is as desired, you can exit the **NUMBER CABLES: PREVIEW OF RESULT** dialog by clicking **OK**. EPLAN finishes the numbering process by writing back the new cable device tags.

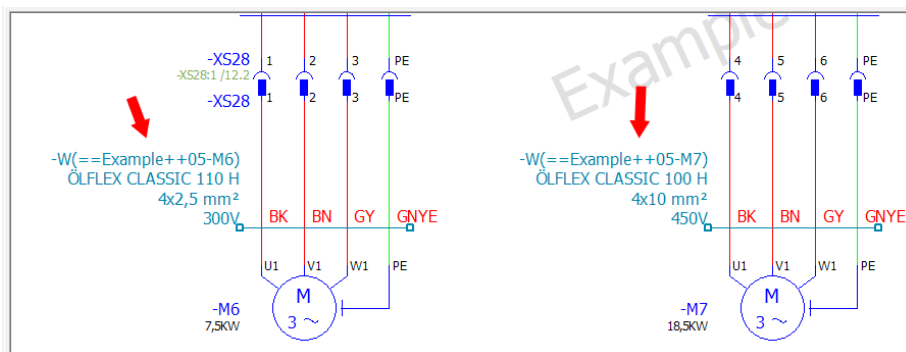


Fig. 5.113 Modified cable device tags

The default settings for the schemes of the different cable editing functions, such as the **Numbering function**, are defined in the project settings. The settings can be found under **OPTIONS/SETTINGS/PROJECT [PROJECT NAME]/DEVICES/CABLE (AUTOMATIC)**.

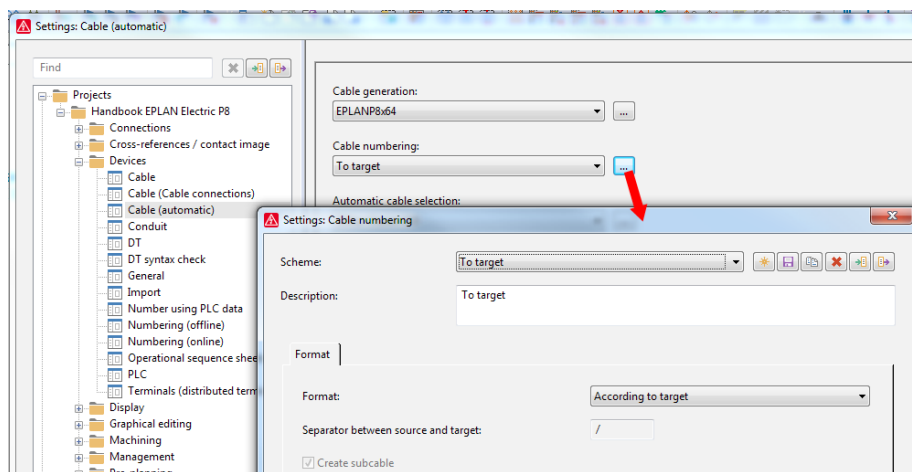


Fig. 5.114 Default settings for cable-editing functions

5.5.3 Automatic cable selection

The **AUTOMATIC CABLE SELECTION** dialog allows you to add a specified cable type to existing cables in the schematic.

To automatically assign cables, in the schematic you again select the cables to be modified and then call up the **AUTOMATIC CABLE SELECTION** function in the **PROJECT DATA/CABLES** menu. EPLAN opens the **AUTOMATIC CABLE SELECTION** dialog.



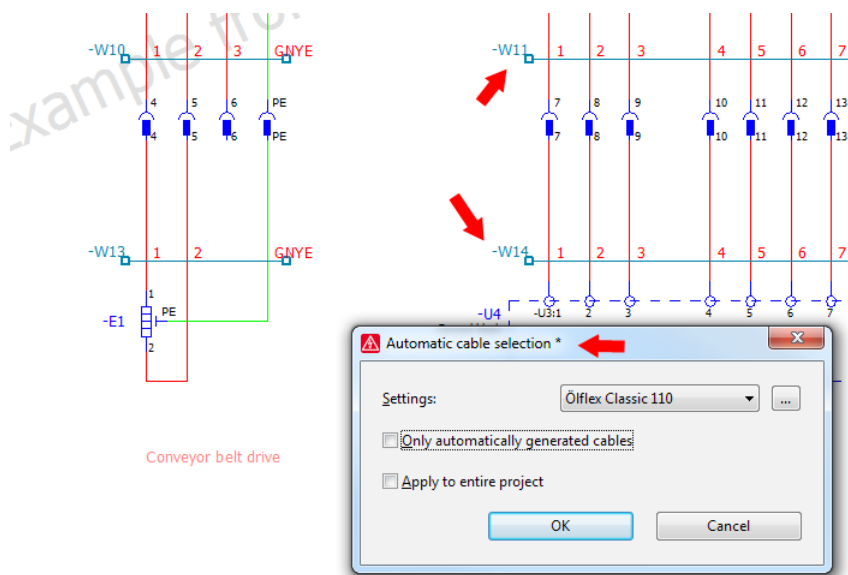


Fig. 5.115 Automatic cable selection dialog

The dialog offers three settings:

- A choice of which cable should be used (an existing scheme can be used or you can create a new one by clicking the [...] button and then selecting it)
- Which cables should be used the automatic cable selection
- Whether or not the automatic cable selection should be applied to the entire project.

The *Scheme for cable types* (settings) can contain multiple cable types. EPLAN automatically determines the required number of conductors and applies this cable to the selected cable DT in the schematic.

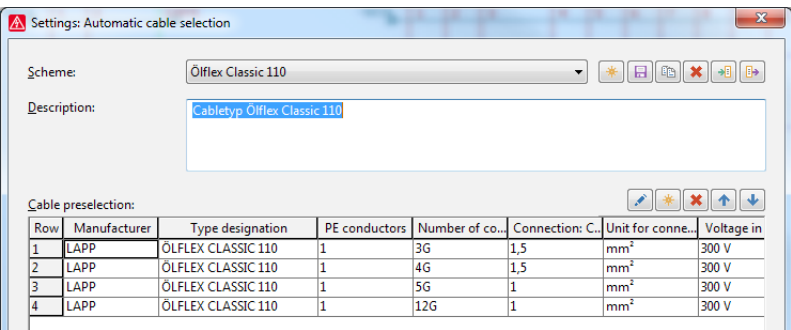


Fig. 5.116 Automatic cable (types) selection settings dialog

Once all settings in the **AUTOMATIC CABLE SELECTION** dialog have been correctly set, you click OK to start automatic cable selection. EPLAN compares the existing cable and

connection data with the entered default cable types and assigns the new cable type(s) to the connections. After this, the cable data is also transferred to the cable definition lines and and/or updated, depending on the settings.

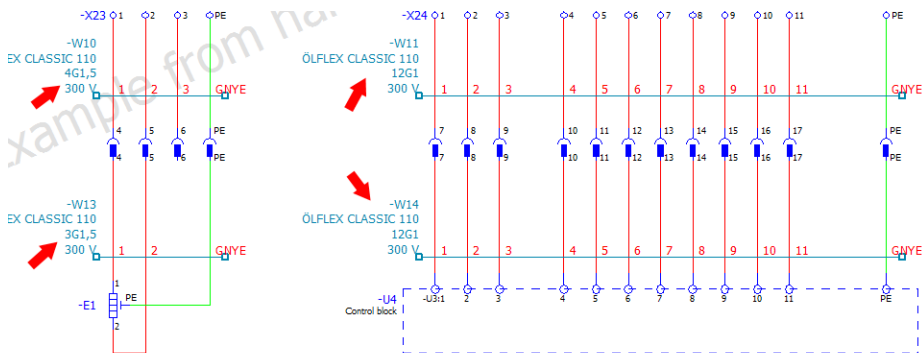


Fig. 5.117 Automatically generated cable data

5.5.4 Generate cables automatically

EPLAN can automatically generate cables and their associated functions such as cable type etc. and thus complete schematics for you.

Preconditions: The cable for which automatic generation is intended already has a cable definition line entered at the connections in the schematic. The cable device tag is adopted exactly from the online numbering. The sequence of the cable device tags plays a subordinate role here. A subordinate role means that the format of the cable device tag can still be defined retroactively in the settings for automatic cable generation.

Before calling up the **GENERATE CABLES AUTOMATICALLY** menu item, you first have to select the corresponding schematic pages in the **page navigator** or the individual cables in the schematic that are to be automatically generated.

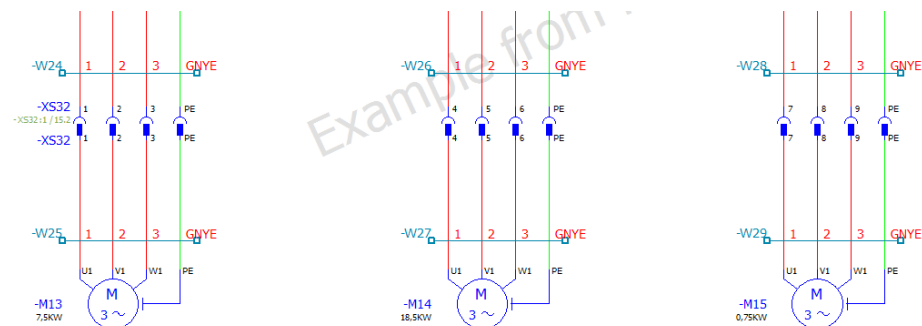


Fig. 5.118 Selecting existing cables

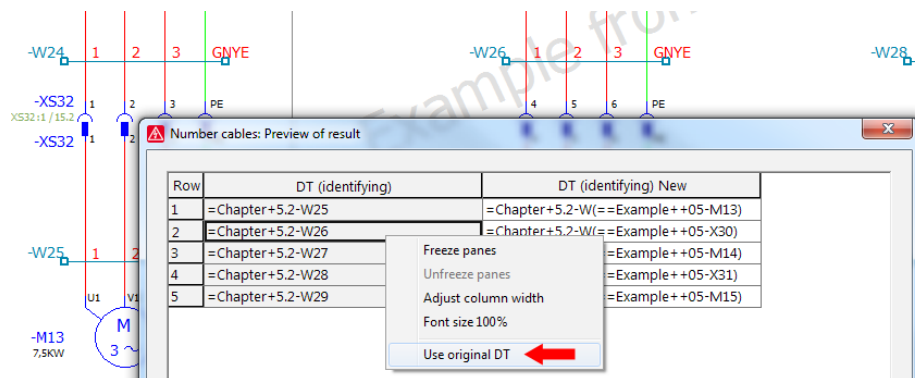


Fig. 5.120 Preview of result

As in other Preview of result dialogs, you can use the popup menu to call the **USE ORIGINAL DT** function. You select the device tag and restore it back to the old DT. It is also possible to manually change the identifier directly in the **DT (identifying) New** field of the preview dialog.

Once all settings are correct, you can leave the **PREVIEW OF RESULT** dialog by clicking **OK**. EPLAN writes the modified cable data back to the selected cable definition lines in the project.

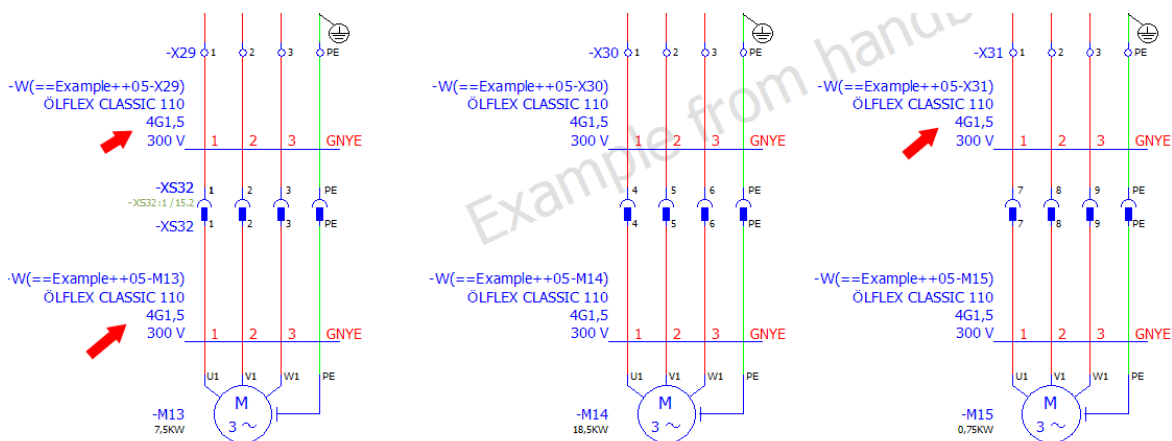


Fig. 5.121 Automatically generated cable data

5.5.5 Assign cable connections

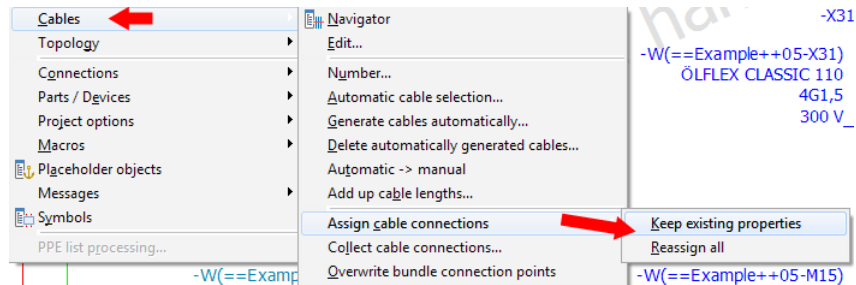


Fig. 5.122 Assign conductors menu item in the Cables menu

In the **PROJECT DATA/CABLES/ASSIGN CABLE CONNECTIONS** menu item, there are two more menu items, the importance and effects of which will be explained in the following sections.

5.5.5.1 Keep existing properties

The **KEEP EXISTING PROPERTIES** menu item enables the new assignment of conductors without affecting conductor assignments that already exist. Accordingly, they keep the cable and conductor assignments as previously assigned. This would be necessary, for example, if cable conductors have already been generated and their assignments should not be changed later. New conductor assignments should be made using this menu item.



All conductors are normally designated during the *Device selection* for the cable. If components, like new cable connections, are to be added from the reserve conductors, the previous conductor assignments should not change. Nevertheless, it is preferable for you to be able to automatically assign the rest of the new conductors to the cable and its connections. This is done via the **KEEP EXISTING PROPERTIES** menu item.

You select the main function of the cable to be edited (to be updated).

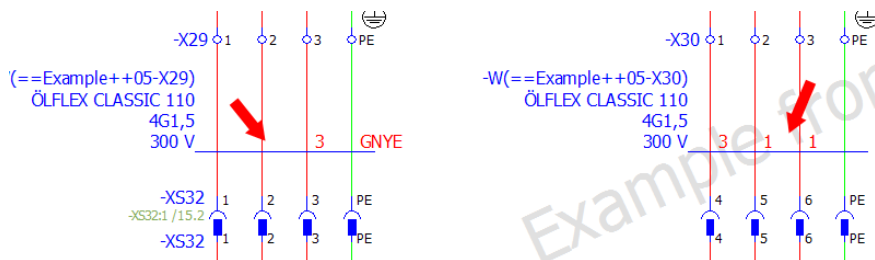


Fig. 5.123 Cables are selected by main function

Now you call up the **KEEP EXISTING PROPERTIES** function via the **PROJECT DATA/CABLES/ASSIGN CONDUCTORS** menu. Based on the function definitions for the conductor designations stored in the cable part, EPLAN writes to the previously unde-

defined connection definition points (conductor designations), whereby existing conductor designations are not overwritten.

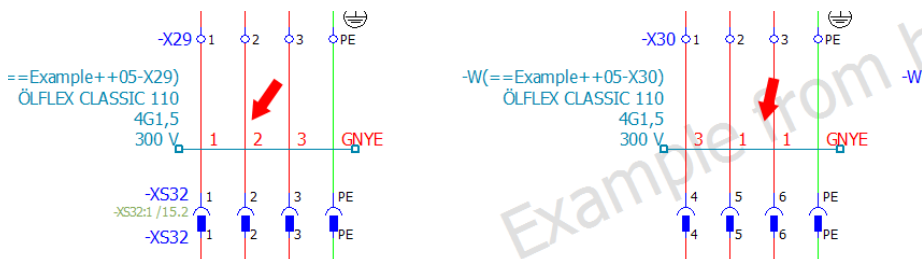


Fig. 5.124 Previously undesigned conductors were given designations.

5.5.5.2 Reassign all

The counterpart to the “Keep existing properties” menu item described in section 5.5.5.1 is the REASSIGN ALL menu item. This function overwrites (rewrites) previously designated conductors with the (conductor) designations from the cable’s function definition. The previous conductor designations are replaced completely. The procedure is as follows.

Select the cables for which the conductor designations should be completely reassigned.

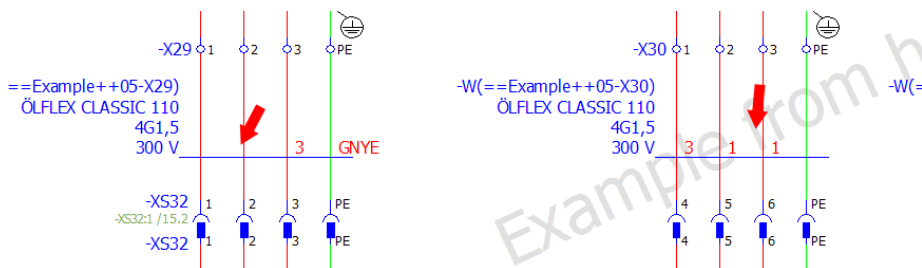


Fig. 5.125 Selecting cables

Next select the REASSIGN ALL menu item from the PROJECT DATA/CABLES/ASSIGN CONDUCTORS menu. EPLAN then reassigns the conductors for all of the selected cables without further prompts.

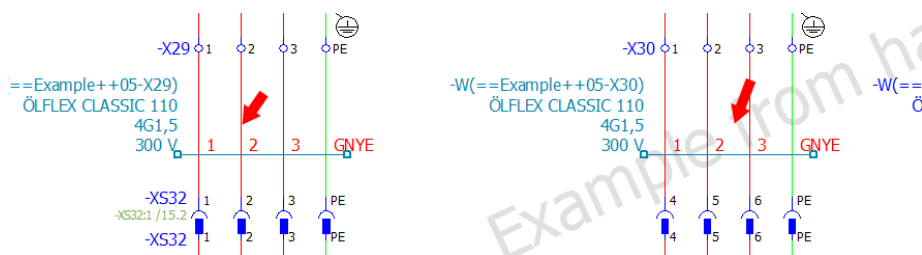


Fig. 5.126 All conductors were reassigned.

5.5.6 New

The **Cable navigator** offers functions similar to those in the other navigators. The **NEW ...** function in the cable navigator popup menu creates a new **CABLE DEFINITION** function.

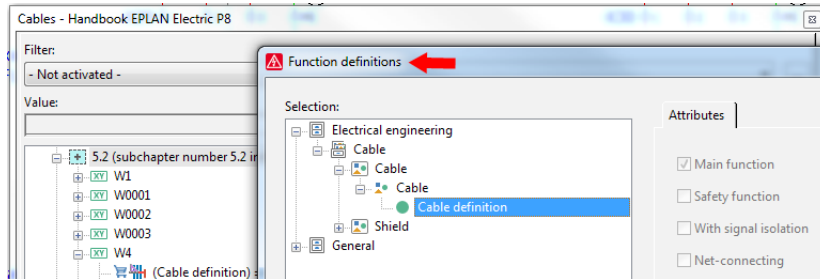


Fig. 5.127 Creating new cable definition

After the import, EPLAN creates a new cable definition and opens the **PROPERTIES (COMPONENTS): CABLE** dialog. Here, the other fields (cable type, etc.) can be filled. If all data has been entered, you can close the dialog by clicking **OK**.

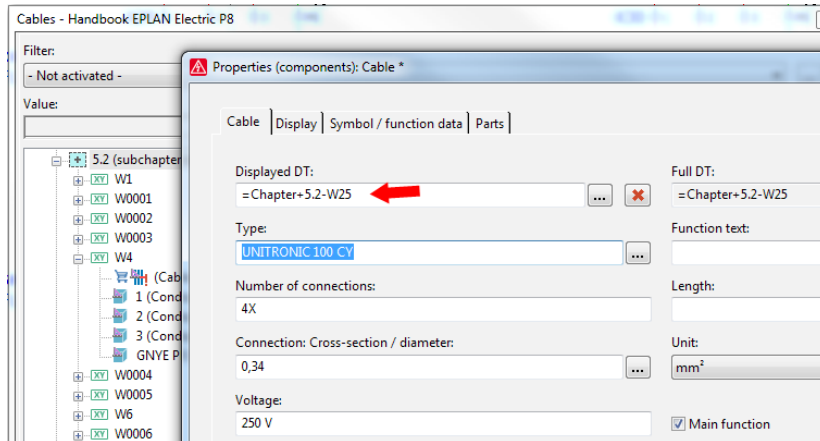


Fig. 5.128 Properties (components): Cable dialog

EPLAN creates the new cable and places it in the structure of the project.

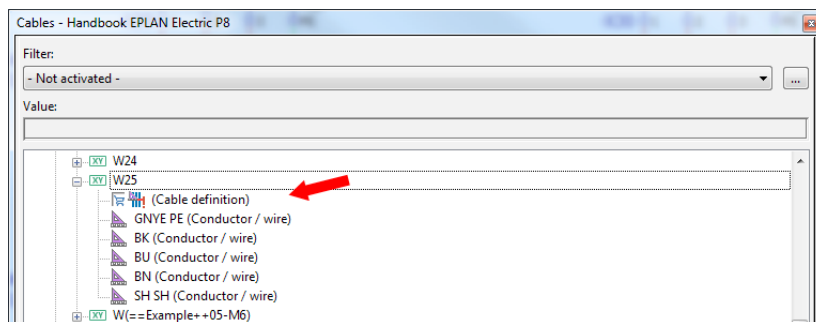


Fig. 5.129 New cable

5.5.7 Number DT

A cable in EPLAN is also a normal device, albeit with special properties. EPLAN not only allows *cable numbering* with, for example, source/target in the **cable DT**, it also allows normal *numbering formats* such as [PAGE, IDENTIFIER, COLUMN].

You select the cables to be numbered in the **cable navigator** or in the schematic, open the popup menu and start the **NUMBER DT** function.

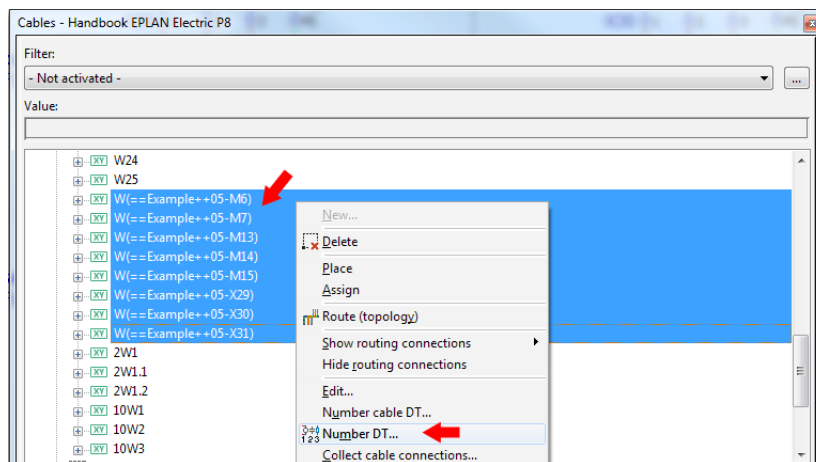


Fig. 5.130 Selected cables in the cable navigator

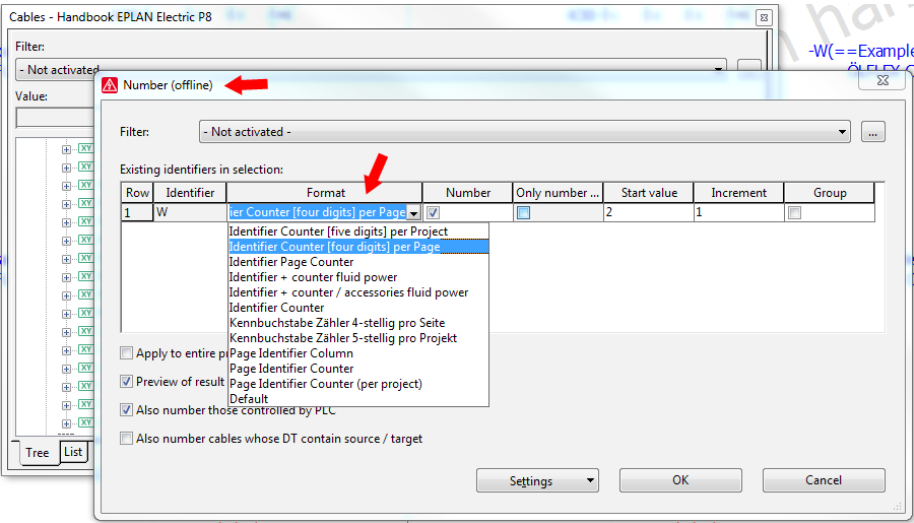


Fig. 5.131 Number (offline) dialog

EPLAN opens the familiar **NUMBER (OFFLINE)** dialog. This is where the desired *numbering scheme* and the other possibly required settings are defined. When you click **OK**, EPLAN numbers the cables like normal devices.

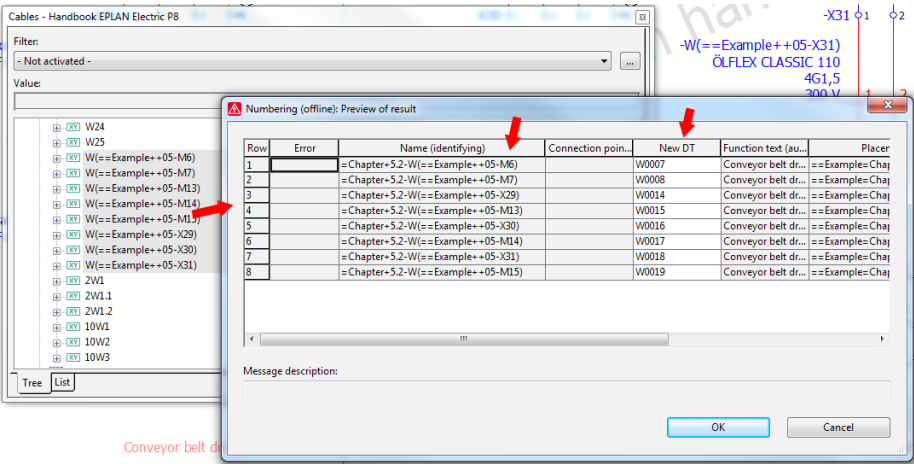


Fig. 5.132 Preview



Fig. 5.133 Final result, numbered cables

■ 5.6 PLC navigator

The **PLC navigator** allows the data of PLC cards and other PLC components used in the project to be managed conveniently from a central place. The PLC navigator shows the individual functions of the PLC cards, such as *addresses*, *power supply connections*, and *channels*.

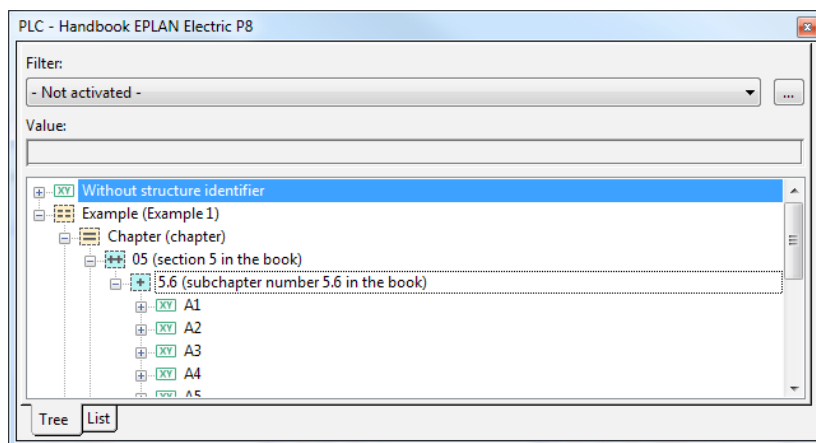


Fig. 5.134 PLC navigator

In addition to the PLC navigator, the **PROJECT DATA/PLC** menu contains other functions for editing PLC data, such as **SET DATA TYPES** or **ADDRESS CARDS**.

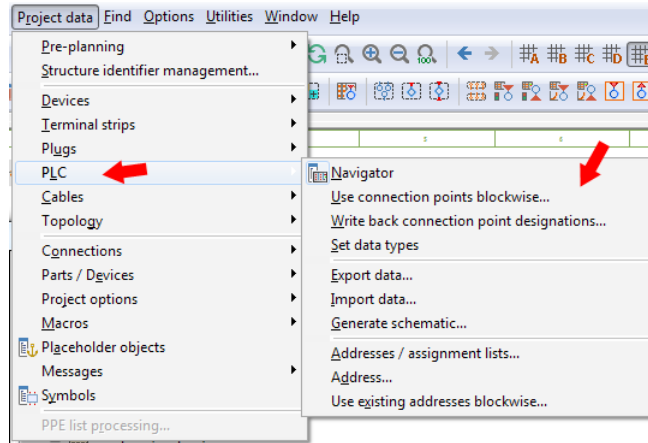


Fig. 5.135 Project data/PLC menu



NOTE: A very important fact is that EPLAN uses the **connection point designation** as the **identifying property** of a PLC function. This means that, for example, the designation E1.0 or A23.7 is not the identifying property, but rather the connection point designation (PIN number or terminal number) of the function of the PLC item.

The representation of PLC items can vary. One option is “dividing by three.” This means that there is a *graphical PLC overview* (rack layout), a *logical PLC overview* (card with cross-references), and the *distributed representation* in the schematic (inputs and outputs used, power supply connections, etc.). This representation is not essential, but it makes it easier to deal with PLC data in a project.

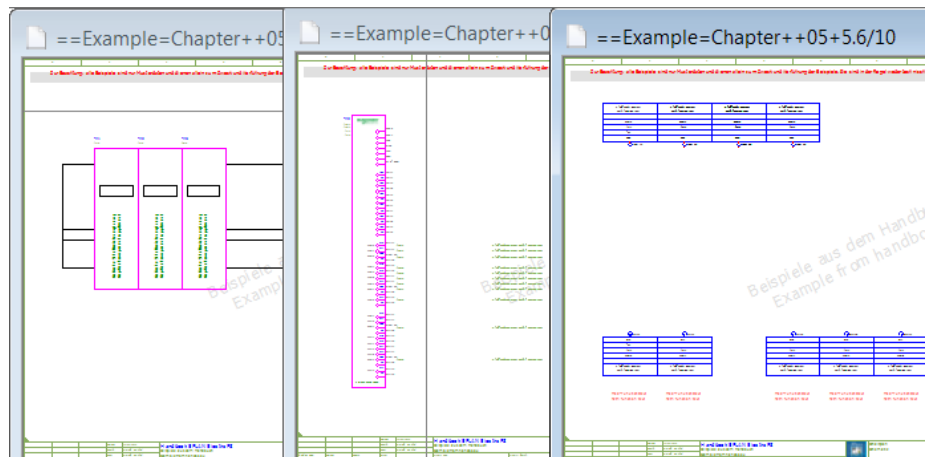


Fig. 5.136 Graphical overview/Logical overview/Distributed representation

The various ways of editing PLC data are described in sections 5.6.1 to 5.6.9, whereby it is assumed that the PLC structure is the same as that described previously, i.e. there is at least one finished logical PLC overview with correct entries (address assignments and connection point designations) and one or more inputs or outputs represented in a distributed manner in the schematic (multi-line display).

5.6.1 Write back connection point descriptions

The **WRITE BACK CONNECTION POINT DESIGNATIONS** function in the **PROJECT DATA/PLC** menu writes the connection point designations from the PLC overview back to the PLC terminals that are represented as distributed in the schematic.

To use the function, you first draw PLC terminals in the schematic or insert them from a macro. The PLC addresses are known and are already correctly entered into the schematic during project editing. The distributed representation of the PLC terminals is finished.

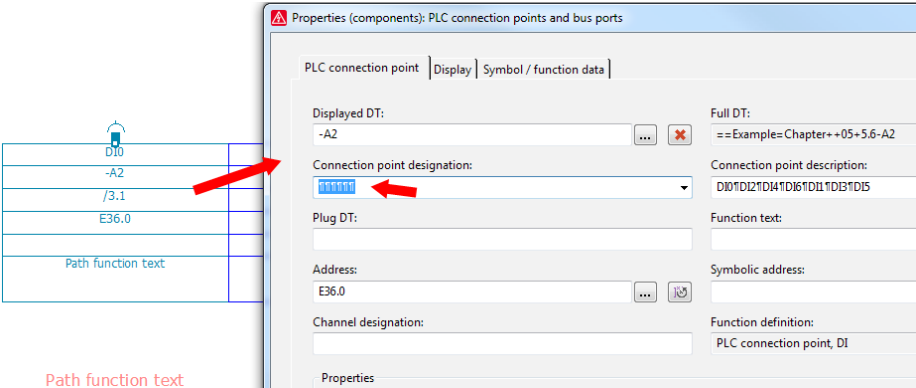


Fig. 5.137 Added distributed PLC data in schematic

When editing is complete, the view is changed to the logical overview page (PLC). Here you select the corresponding card(s) and address ranges. Then you select the **WRITE BACK CONNECTION POINT DESIGNATIONS** item in the **PROJECT DATA/PLC** menu.

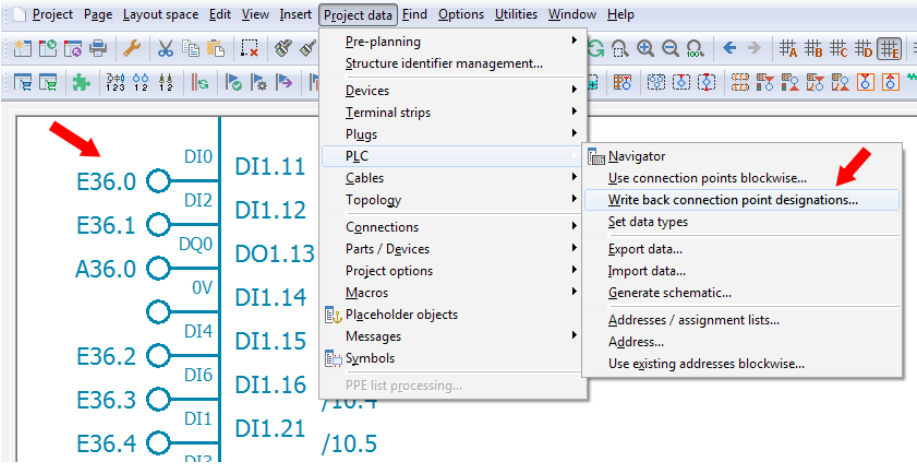


Fig. 5.138 Running the Write back connection point designations function

EPLAN opens the **WRITE BACK CONNECTION POINT DESIGNATIONS** dialog. Here, select the desired CPU and appropriate options. You confirm this selection by clicking OK.

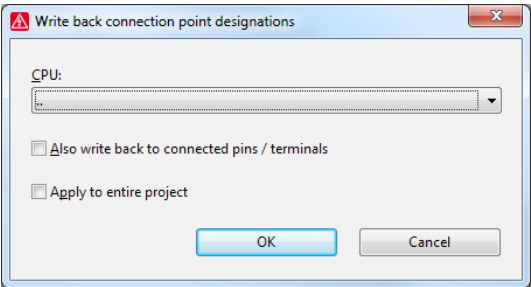


Fig. 5.139 Write back connection point designations dialog

The connection point designations from the logical PLC overview (of the selected PLC terminals) in the schematic are now written back to the PLC data that is in distributed view.

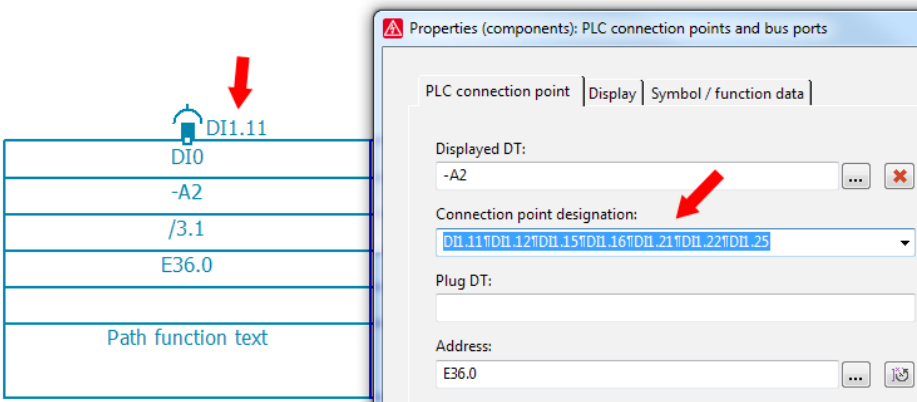


Fig. 5.140 Written-back connection point designations

The result can be seen in Fig. 5.140. All connection point designations are correctly transferred to the PLC terminals. But this only works if the addresses themselves have been entered correctly. Otherwise, EPLAN cannot recognize the relationship between the PLC terminal in the logical PLC overview and the PLC terminal in the distributed view.

5.6.2 Set data types

The **SET DATA TYPES** function is used to automatically update and write the corresponding data types to the PLC terminals if they are not already there. EPLAN derives the data types from the functions. You select the PLC terminals in the *logical PLC overview* or the *PLC terminals in the schematic* and call the **SET DATA TYPES** function in the **PROJECT DATA/PLC** menu. EPLAN writes the data type into the properties of the PLC terminal.

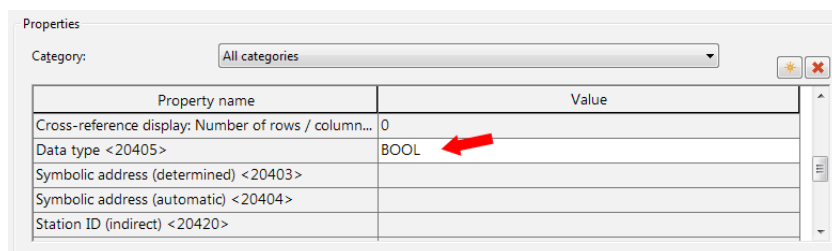


Fig. 5.141 Data type set

5.6.3 Export/Import data

This menu item allows for PLC data to be exported/imported after various PLC controls.

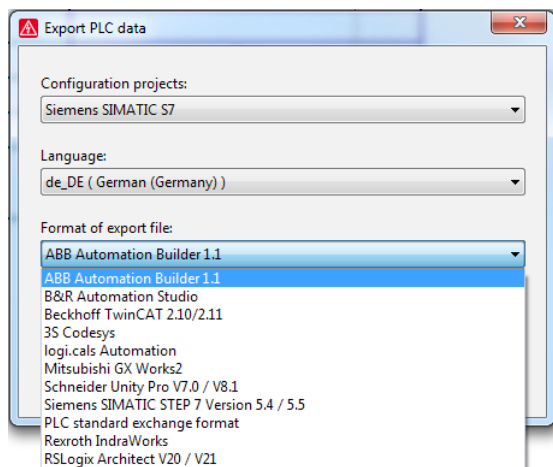


Fig. 5.142
Export PLC data dialog

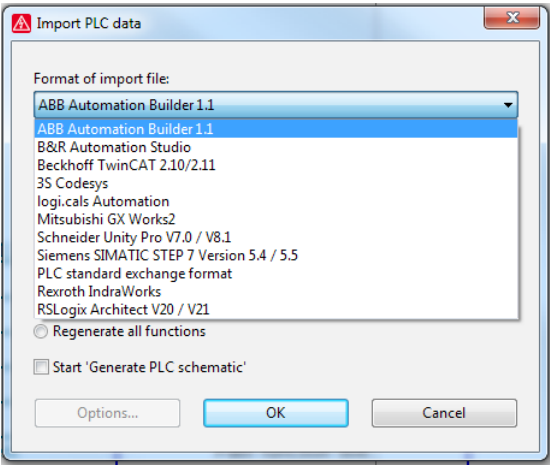


Fig. 5.143
Import PLC data dialog

5.6.4 Addresses/assignment lists

In the PROJECT DATA/PLC menu, the ADDRESSES/ASSIGNMENT LISTS function opens the dialog of the same name (a table of all addresses and CPUs used in the diagram).

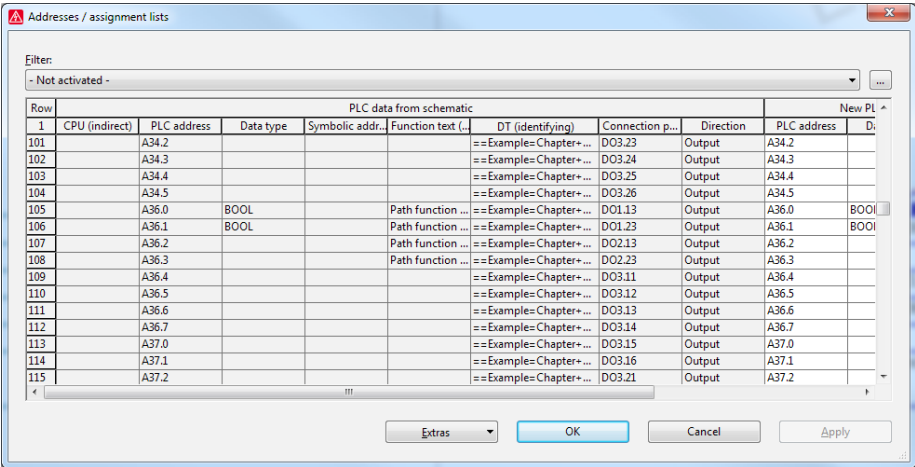


Fig. 5.144 Addresses/assignment lists dialog

In this dialog, the properties like the *PLC address*, the *Symbolic address* and the defined *Function text* of the respective CPU area selected can also be changed manually. EPLAN then automatically writes back the modified data to the PLC terminals and the modified function text to the manually placed path function text in the PLC terminal path. At this point, if you remain at the function texts, it is also possible to edit the function texts so that they can be written back to the schematic. Since this is done at a central place, it

avoids tedious paging through the schematic in order to modify the placed path function text at the corresponding PLC terminals.

In addition, an *assignment list* can be exported from this dialog (for the downstream PLC software) or imported (from the PLC software). But before such an action, the CPU must be defined via a filter. If this has not been done, EPLAN will display a message.

CREATE FILTER FOR CPU can be selected from the popup menu. EPLAN reads out the CPUs and creates a suitable filter automatically.

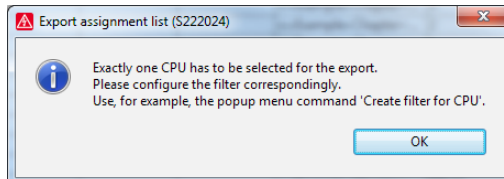


Fig. 5.145
Message about missing
CPU filter

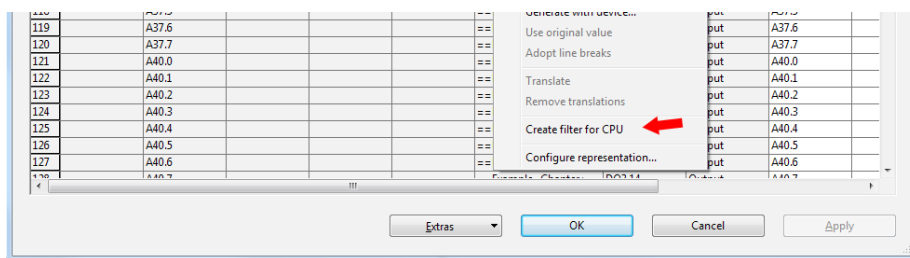


Fig. 5.146 Popup menu with Create filter for CPU command

After clicking it, EPLAN creates the filter.

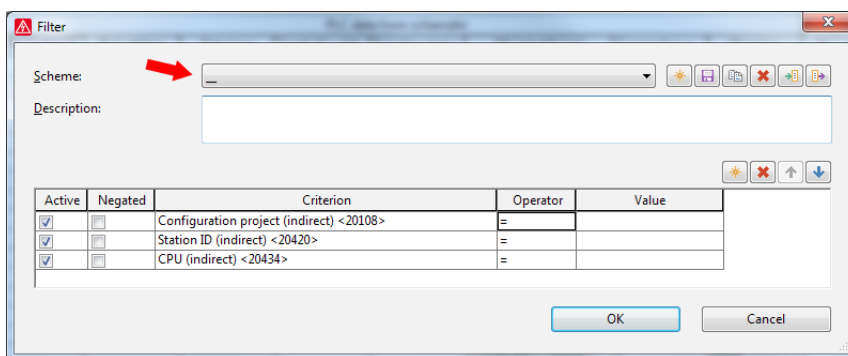


Fig. 5.147 Automatic filter generated by EPLAN

Click the **EXTRAS** button in the lower area of the dialog, and EPLAN opens another menu.

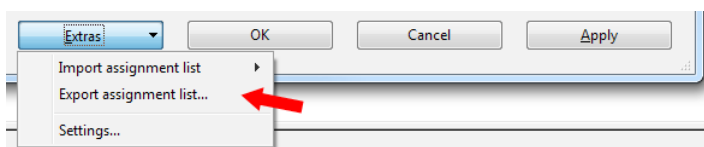


Fig. 5.148 The Extras button and its functions

This allows *assignment lists* to be *exported* or *imported*, including new CPU assignment lists. Different settings may be required, depending on the PLC and its software.

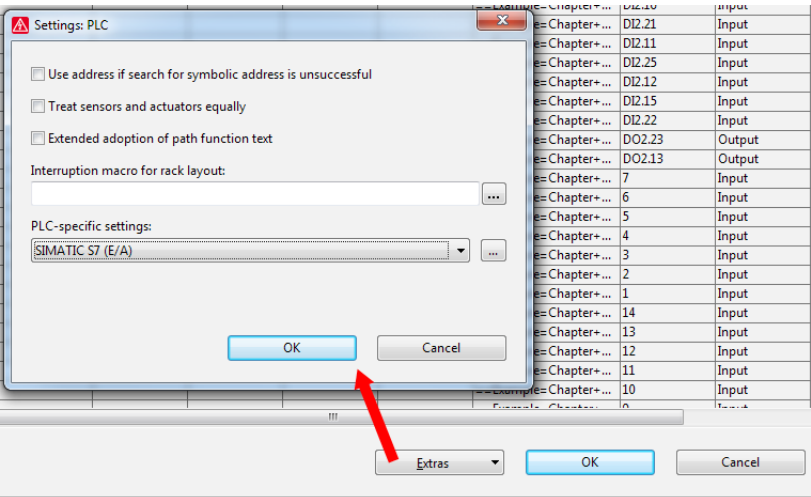


Fig. 5.149
PLC settings

EPLAN already contains predefined schemes for a range of PLC software for export and import, so that all you need to do is select the correct PLC software in the settings under *PLC-specific settings*.

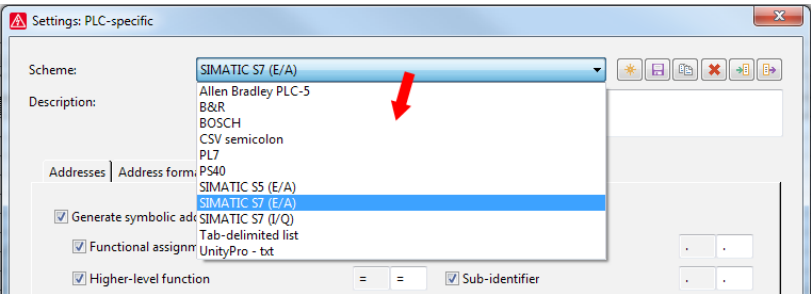


Fig. 5.150
Various PLC schemes

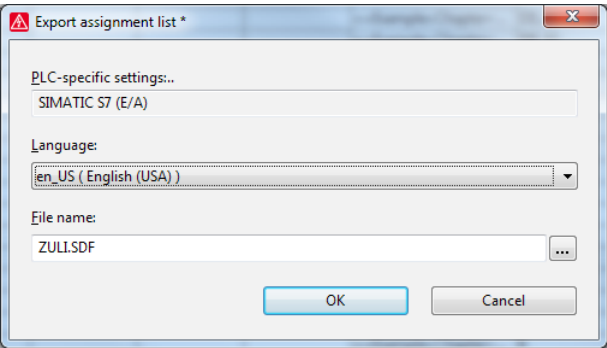


Fig. 5.151
Export assignment lists dialog